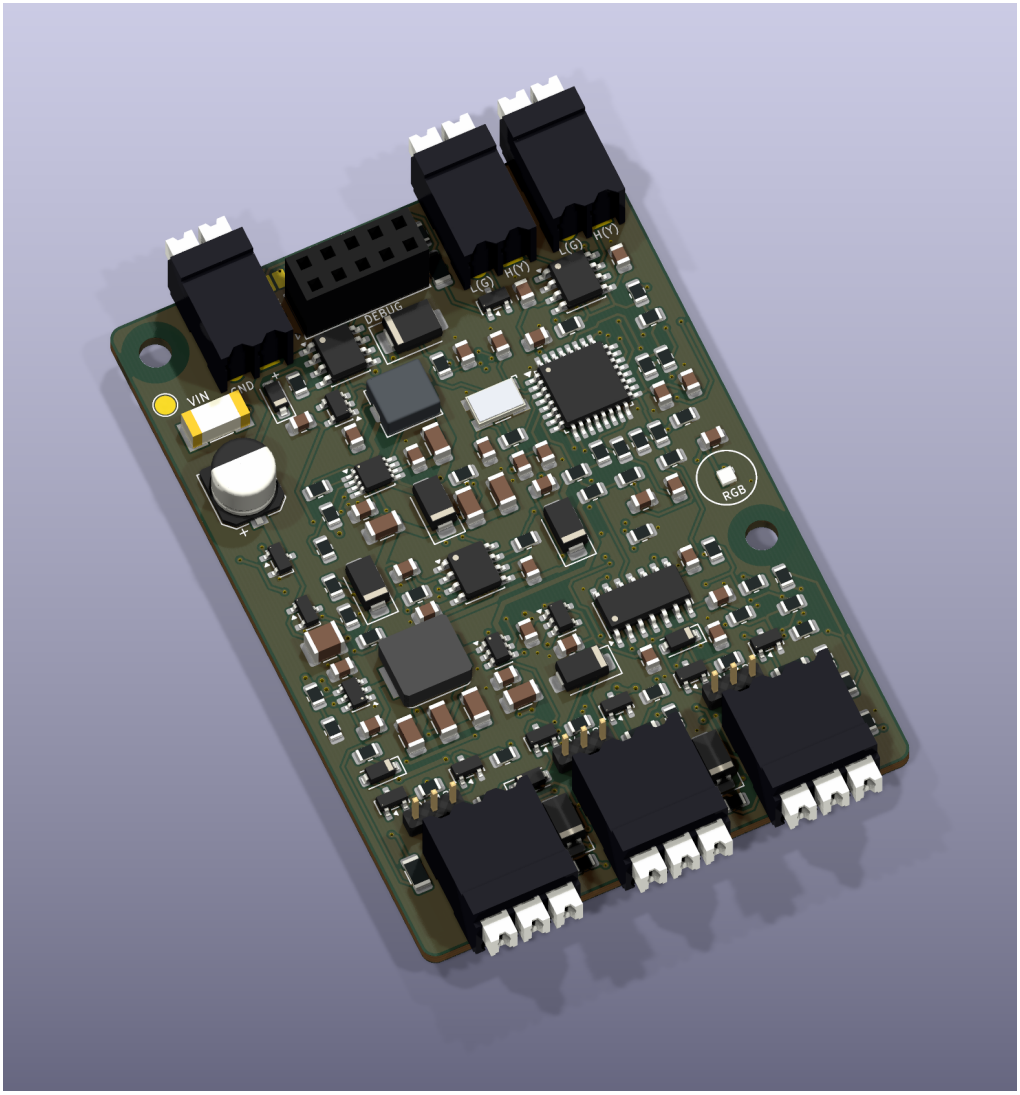
sugarCANe

Revision H SC12 · Prototype engineering and assembly guide

Three industrial / logic sensor inputs
Regulated 12 V and 5 V sensor power



2026-09-10

Rev H adds RGB diagnostics and inexpensive header programming while retaining protected sensor power and physical voltage jumpers. No physical prototype has been tested. This guide accompanies the engineering evidence and defines first-article acceptance.

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1 Product overview

Source: README.md

Repository: barkinvar/sugerCANe · Folder map

Three digital sensor inputs over CAN, with regulated **5 V / 12 V sensor power** and software-selectable **NPN, PNP, push-pull, floating detection and keeper**. Removable 2 mm jumpers select each port's power voltage. All components mount on the top.

The bold pixel **sugarCANe** wordmark and original pixel-cane illustration on a raised island with a stepped underside fill the underside; it carries no other text. Connector and voltage labels remain on top, with the full programming map in the wiring guide. Bottom render · Vector artwork and lettering source.

Rev H adds one enclosure-facing RGB status LED and a hand-soldered recessed female programming socket for a ten-board batch. The LED is on the right, above the mounting hole, with a 6 mm circular area reserved for a printed light duct; see the enclosure dimensions. It retains Rev G's power protection and physical voltage jumpers. The 12 V reservoir is upstream of a controlled output eFuse; both rails have default-off enables and CAN fault reset. Two ADC channels measure the protected rails. Engineering review · verification and bench procedure. This design uses its own CAN protocol; it is not CTRE Phoenix or GrappleHook compatible.

Selected assembly plan: JLC fits **129 top-side SMT parts**. You buy **STM32F042K6T6 from DigiKey** and solder U5, then fit ten through-hole parts and three jumper caps. U5 is omitted from the factory BOM, CPL and stencil. All 54 factory types had stock for ten boards in the September 10, 07:22 UTC check. Full power/input/firmware and four-layer CAD screens pass; factory process/stack/inspection confirmation and quotation remain pending. **Start here: where to buy and how to order.**

Specification	Rev H
Board supply	6-22 V DC
Sensor supplies	Regulated 5 V / 12 V; **300 mA shared per rail**, not per port
Signal targets	Positive 0-30 V; LOW ≤1.2 V, HIGH ≥2.4 V; common ground
Input setup	Software bias and active-state selection independently per channel
Protection	Reverse input, fuse/TVS, separate protected 5 V and 12 V outputs, controlled 12 V inrush
Fault reset	Remove the fault, then use CAN power reset; hardware limits remain independent
Controller	STM32F042K6T6, TCAN3413; 1 Mbit/s classic CAN
Software	App 0.9.0, bootloader 1.1.0, hardware 8, protocol 4
Servicing	CAN discovery, ID/settings and recoverable firmware upload; initial installation through a 2.54 mm SWD service socket
PCB / assembly	**52 × 74 mm**, four copper layers, 129 factory SMT + user-fitted MCU + ten through-hole parts + three caps

Parts purchasing is about **$27.18 per board at ten boards before SMT attrition**, or **$28.06** including a conservative allowance. These include the user-fitted DigiKey MCU, through-hole parts and caps; two optional spare MCUs are extra. PCB, assembly, freight and tax are extra. Cost calculation.

Change voltage jumpers only with power disconnected. A missing cap disconnects port power, not its signal wire. Input mode changes use CAN software. Rail voltage readback uses the MCU ADC and factory voltage-reference calibration. External signal inputs remain digital; rail measurement does not add user analog-input or encoder support.

The 52 × 74 mm board has top-side assembly, rounded corners, improved edge clearances, a recessed debug socket and labelled voltage selectors. Its 1,530 straight track segments and 242 vias satisfy the routing rules. All-layer cleanup removes pad overshoots, redundant paths and stubs, with explicit chamfered branches. Ground pours use checked widths and gaps; see the ground cleanup. Component positions and artwork are unchanged. See before/after details, all four copper layers and connector details.

The input connector-to-fuse and fuse-to-MOSFET routes are now **0.7 mm wide** (14 segments widened from 0.5 mm). The protected input and converter paths retain their reviewed clearance necks. Width, clearance, connectivity and ground-pour checks cover **all four layers**; see the power-path review.

Copper layer	Role	PNG	SVG
Top	Components, signals and power	PNG	SVG
Inner 1	Ground plane	PNG	SVG
Inner 2	Ground with selected signals and low-current supply branches	PNG	SVG
Bottom	Signals, power and ground	PNG	SVG

These views show the actual exported copper and drills. Inner layers are viewed from the top; the bottom is mirrored for an underside view.

2 Pre-order decision

Source: docs/ORDER_READINESS.md

Engineering screens: PASS. Manufacturing order: HOLD pending factory confirmation. Final pre-order verification is complete against the selected DigiKey-MCU/JLC assembly BOM. C25/C26 use 20 pF clock-load capacitors; U1/L1 now use a 3 A LMR51430XDDCR and 8.2 µH SRP7028A-8R2M in unchanged footprints. The C1 high-reflow replacement is retained. Native CAD, all-layer connectivity, BOM matching, input simulations and firmware checks pass. Both complete coupled-startup runs pass. Factory process/stack/inspection confirmation and quotation remain open. No physical board has been tested.

The current BOM passes **nine power cases plus a complete startup numerical repeat, ten input cases and sixteen firmware waveform replays**. Both startup traces reach 40 ms. Protected 12 V settles at 11.975-11.980 V, peaks at 12.054 V and draws a peak 1.544 A through L2. The full-run comparison differs by at most 10.37 mV at the protected output and 0.524 mA in peak inductor current, within the unchanged 50 mV / 20 mA limits. The largest compared rail-voltage difference is 14.23 mV. Current power report · Waveforms.

The native routed-copper calculation was rerun after widening 14 input-route segments from 0.5 to 0.7 mm and again after the C1 footprint replacement. At the stated 85°C/nominal-copper assumptions, the buck-output-to-boost-input drop remains 51.14 mV against a 120 mV allowance. The worst protected-supply-to-jumper branch drops 42.49 mV at 300 mA; including jumper-to-connector copper brings it to **44.01 mV against the existing 60 mV allowance**. Each rail has a shared 300 mA sensor budget; this tests that entire load at each port in turn.

The DC model now recognizes overlapping track copper, contacts inside a straight segment and via connections on multiple layers, using the saved stackup depths. Buck-to-boost drop is measured between those two terminals; the older value was relative to the buck feedback tap. These figures therefore are not a direct before/after performance comparison. Six resistor-network regression fixtures check known resistance, segment subdivision, contacts, via depths, voltage-reference independence and genuine opens. The model still assumes ideal pad/via-annulus copper and does not solve current crowding or heating.

Routing cleanup uses 1,530 segments and 242 vias. The routing gate rejects pad overshoots, overlapping routes, tiny free jogs, end-cap-only joins, sharp free bends, unchamfered branches and unreviewed widths. All 77 routed nets satisfy the updated width policy. The ground-pour pass adds 0.25 mm width/gap rules, solid SMT ground connections, five local cutouts and a corrected J8.7 thermal-spoke angle. Three redundant J6.2 ground segments were removed; that pass preserved signal/power routes and all 242 vias. The input pass changed only 14 widths and regenerated pours; the subsequent C1 substitution preserves all 1,530 traces and 242 vias. Local converter/crystal path limits still pass. Fresh native DRC and independent reconstruction of all four exported copper layers find no shorts or opens; see the routing results and four-layer copper review. These checks do not replace physical qualification.

The input screen uses 2 A through J1 → F1 → Q1 → U1, with finite via resistance. It is a DC screening point above the normal analytical 1.726 A at 6 V, not a fault-current or thermal rating.

Input copper path	Width	Drop at 2 A, before → current
J1 to F1 (VIN_RAW)	0.5 → 0.7 mm	27.42 → 19.03 mV
F1 to Q1 (VIN_FUSED)	0.5 → 0.7 mm	41.14 → 30.90 mV
Q1 to U1 VIN (VIN_PROT)	0.5 mm trunk / 0.8 mm local bypass	91.44 → 90.47 mV
Total input feed copper	All three paths	**160.00 → 140.40 mV**

The table compares the design before input widening with the current C1 lands. The small VIN_PROT difference comes from the changed ideal-pad contact representation; it is not evidence of improved thermal performance. The input copper resistance falls about 12%, to 70.20 mΩ. At the normal analytical input current, its drop is approximately 121 mV. The existing 276 mΩ entry-resistance planning budget therefore leaves approximately 206 mΩ for the excluded fuse, MOSFET, contacts, cables and ground return; those items still require verification. The protected-input trunk cannot simply be widened in place without violating clearances near J1 and D3. The low-current U1 enable branch remains 0.25 mm.

All four layers retain nominal **35 µm (1 oz) copper**, now explicitly enforced by the DC screen. Inner 1 is the ground plane; inner 2 has 92 selected signal/service/control and low-current supply segments alongside

ground. Native routing rules and independent exported geometry checks include both inner layers. Individual PNG/SVG views and the four-page copper PDF are regenerated from the delivered Gerbers.

Final component-margin review

C25/C26 changed from 30 pF to **20 pF C0G / Yageo CC0805JRNPO9BN200** without moving pads or traces. The conservative crystal gain calculation improves from 4.61 to **6.54** at 10 pF assumed stray capacitance and the published worst component bounds. All six modeled loading corners exceed ST's gain-margin recommendation. Frequency pulling, oscillator drive, actual stray capacitance and startup must still be measured on the first board. Calculation and assumptions.

The buck review found a second margin issue: the former 2 A LMR51420's minimum 1.7 A valley limit can be exceeded by maximum 12 V charging current plus a normal 5 V load near minimum input, as well as by some simultaneous-fault corners. **U1 is now LMR51430XDDCR and L1 is SRP7028A-8R2M, 8.2 µH**, in the same footprints. The new minimum limits are 2.75 A valley and 3.67 A peak. All nine normal/charging/fault current-limit corners pass; the maximum calculated peak is 2.625 A. L1's published 4 A RMS and 7.5 A saturation ratings exceed the modeled load and the regulator's 6.68 A maximum peak limit. This improves current headroom; dropout, supply impedance, thermal behavior and uninterrupted MCU/CAN operation still require bench testing. Calculation · Exact change scope.

The public TI LMR51430 product page does not provide a compatible switching model. Its upstream control loop is **not simulated** here. All ten completed downstream power decks match final-BOM generation byte for byte because U1/L1 lie upstream of their idealized source. That preserves the downstream evidence, without treating it as a simulation of the new buck.

Changes that address the earlier failures

Rev F finding	Retained power implementation	Verification
Low-input charging current exceeds former buck minimum current-limit headroom	3 A LMR51430 with 8.2 µH / 7.5 A-saturation L1	All nine current-limit corners pass; upstream loop/dropout/thermal qualification remains physical
Input-side limiter interrupts loaded boost startup	U8 connects to internal 5 V; TPS259620 output eFuse connects the sensor load after the raw boost reservoir	Both complete Rev H startup runs pass with 66 µF raw reservoir, 100 µF external load and 300 mA
Reservoir discharge passes through the boost short-current path	U9 limits the protected output; C62-C64 remain upstream; D5 clamps negative output excursions	Running short, cold short, sustained fault and restart scenarios
Firmware cannot reset power protection	PA15 controls U4; PB7 controls Q3/Q2 and U9 enable	Register-level pin tests, application sequencing and addressed CAN reset tests
Reset/unpowered MCU needs a safe output state	R71/R75 pull enable signals LOW; R70 derives the 12 V inhibit from raw BOOST_12V	Native pin/net checks and default-off circuit review
Coarse rail-presence indication misses undervoltage	Separate 11:1 dividers and RC filters feed PB0/PB1 ADC; firmware uses calibrated internal reference	ADC register tests, rail-state tests and divider/current budgets
Local boost bypass below recommended value	C61 increased to 1 µF; 22 µF bulk C60 retained	Exact-MPN BOM/model/parity checks
Buck input bypass left remote from U1	C2/C3 moved beside VIN; explicit local top-side ground return, with bootstrap routing clear of it	Native path checks require C3→VIN below 2.2 mm and C2→VIN below 5.5 mm; fresh DRC and Gerber connectivity

The 5 V/12 V selection remains a removable 2 mm cap on each three-pin header. All 140 fitted parts remain on top: 129 factory SMT placements plus the user-fitted MCU and ten through-hole parts. The PCB is 52 × 74 mm, four copper layers.

Evidence to read

Release check index, electrical budgets, native-route DC calculation and pin review, 768-corner averaged boost-loop screen, ten input-transient cases, nine power cases and startup repeat, firmware waveform replay, and full power waveforms. Earlier Rev G evidence remains in the archive.

The DC calculation uses nominal 35 µm copper at 85°C and 20 µm via plating, with ideal pad contacts. It checks the 120 mV boost-input PCB allowance at a conservative 1.3 A boost branch, 510 mA 5 V branch, 130 mA logic load and 20 mA RGB/buffer load. Ground return, connectors, cables, actual copper tolerances and temperature rise need measurement.

Pin expectations are checked independently against manufacturer functions for the MCU, CAN transceiver, comparators and power switches. Native ERC/DRC, model/schematic/PCB parity and reconstructed Gerber connectivity check the actual delivered design rather than only the conceptual schematic.

Limits and first-article acceptance

The vendor transient model uses an idealized upstream 5 V source; it does not simulate the LMR51430 loop, robot wiring, all silicon corners or an assembled board. The eFuse's floating OVCSEL setting is a roughly 13.8 V fault clamp, not a precision 13 V trip. Software rail monitoring supplements the independent hardware current limits. It does not make a fault-proof or reverse-powered output.

The passing short cases are **shorts to ground**. Directly joining the 5 V and 12 V rails is unsupported and can damage the 5 V switch or sensors: U4's OUT absolute maximum is 7 V. Its reverse-voltage blocking and software overvoltage detection do not guarantee survival of a 12 V cross-connection. See the wiring limits.

Qualify startup, both rails loaded together, the exact sensor models and cables, brownouts, cold/running shorts, temperature and CAN programming/recovery using the validation procedure. Record PCB revision H, exact BOM and firmware identity. Simulation and manufacturing checks reduce the risk of a first order; they cannot guarantee first-board success.

Primary pin/application references: STM32F042, TPS2596, TPS61085, TPS2553, TCAN3413, LM339LV.

3 Power and input circuit

Source: DESIGN_PLAN.md

Rev H retains G's protected power circuit and adds RGB status and ordinary-header programming. The regression report records every change against the frozen G circuit.

Supply path

```
6-22 V → fuse / reverse protection / TVS → internal 5 V, 3 A buck
                                          ├─ 3.3 V MCU / CAN / comparators
                                          ├─ U4 latch limiter → SENSOR_5V
                                          └─ U8 boost → reservoir → U9 eFuse → SENSOR_12V
SENSOR_5V / SENSOR_12V → removable jumper per port → sensor power
```

U1 is LMR51430XDDCR, paired with L1 SRP7028A-8R2M (8.2 µH). The final current-margin review replaces the former 2 A buck and 10 µH inductor in the same footprints; the 300 mA shared sensor-rail limits are unchanged. U8 remains TPS61085DGKR, with the 3.3 µH shielded inductor, B240A diode and 13 kΩ / 33 nF compensation. C61 is now a 1 µF local bypass. C60 remains 22 µF, and C62-C64 remain three 22 µF reservoir capacitors. Their effective capacitance at 12 V is much smaller than their nameplate value; the checks include both explicit derated and nameplate cases.

A boost has a direct path through its inductor and diode. Rev H puts **TPS259620DDAR after the reservoir**, so a sensor short cannot discharge that reservoir directly through the port. R62=2.61 kΩ sets about 357 mA nominal output current limit; the tolerance budget is in electrical_checks.json. Useful sensor load remains **300 mA shared per rail**. C70=47 nF controls output slew; C65/C71 provide local input/output bypassing. D5 clamps negative output transients and D6 limits raw-rail transients. These parts do not authorize reverse powering a sensor port.

U9's UVLO divider requires approximately 11.1 V on the raw boost before connection. OVCSEL is intentionally open for its **13.8 V-class fault clamp**, not a precision 12 V ceiling. R70 pulls Q2's gate from raw 12 V, holding UVLO low by default. PB7 HIGH enables Q3, which removes that inhibit. R75 holds Q3 off without MCU power. U4's PA15 enable has a hardware pulldown. Both branches therefore start disabled; their current limits operate independently of firmware.

Firmware starts 5 V after 10 ms and 12 V after 60 ms. It allows 200 ms for startup, requires 100 ms healthy voltage before counting inputs, and latches a running low-voltage fault after 20 ms. Hardware fault indications or excessive measured voltage disable the affected rail. Removing a short does not automatically restart it: use the explicit CAN power-reset command. The bootloader keeps both rails off.

Rail measurement and signals

PB0/ADC8 and PB1/ADC9 measure the protected 5 V and 12 V rails. Each divider uses two 100 kΩ precision series resistors, a 20 kΩ bottom resistor and a 10 nF filter. Firmware uses 239.5-cycle acquisition and the factory VREFINT calibration value. Readback and rail states are available through CAN service. This measures the board rails, not the voltage at a remote sensor.

The former coarse fourth comparator is unused, with defined inputs and an unconnected output. The three signal comparators and electronic NPN/PNP/push-pull/float/keeper circuits retain their original values. Two pulldown controls move to PB5/PB6; channel 3 stays on PB4. Selecting 12 V sensor power never changes the signal pull-up from 3.3 V.

External inputs remain digital: target LOW ≤1.2 V, HIGH ≥2.4 V, positive 0-30 V and common ground. Sensor off-state leakage, cable capacitance and residual voltage must meet the input-mode limits. Rail ADCs do not add external analog or encoder support.

Layout and assembly

The board is 52 × 74 mm with four copper layers and all 140 fitted components on top. The factory places 129 SMT parts; the user hand solders the DigiKey STM32F042K6T6 MCU, six terminals, three voltage headers and one J8 service socket, then fits three loose caps. U9 uses a split-paste exposed pad and surrounding thermal vias. No blind/buried vias or second-side assembly is required.

One inner layer provides continuous ground; the other also carries a few slow control/service connections. Switching, feedback, compensation and crystal routes remain on top. Native and independently reconstructed fabrication geometry are checked separately.

Use hardware 8, application 0.9.0, bootloader 1.1.0 and protocol 4 from the same release. Verification distinguishes calculations and simulation from physical qualification. Cost basis includes supplier minimums and excludes assembly/freight/tax.

Rev H design envelope: at most 100 µF total external capacitance on the shared 12 V rail, and 22 µF on the shared 5 V rail, at 300 mA per rail. The startup calculation includes +20% capacitance and the current-limit minimum; the 470 µF simulation is a typical diagnostic beyond that envelope. Qualify nonlinear sensor inrush separately.

RGB diagnostics and low-volume programming

H replaces the single green LED with one SK6805-EC15, a 5 V AHCT buffer, data pulldown, series resistor and local bypass capacitors. PB3 remains the status pin and uses timer/DMA serialization. J8 is a hand-fitted recessed female 2×5 2.54 mm socket with the same electrical pin numbers and odd/even physical pattern as the preceding Rev H header. Status, case opening and inexpensive programmer.

4 Wiring and programming header

Source: docs/WIRING.md

Use the included hardware-8/protocol-v4 firmware and tools; older-revision firmware must not be used on Rev H. Board coordinates below are viewed from the component side, origin at the top-left PCB corner. Bottom silkscreen is read after turning the board over. Do not infer pin numbering from left/right after flipping.

Connector	Pin 1	Pin 2	Pin 3
J1 power	Positive input, 6-22 V	GND	-
J2, J3 CAN	CAN H, yellow: H(Y)	CAN L, green: L(G)	-
J4 input 1	Selected sensor power	GND	Signal
J5 input 2	Selected sensor power	GND	Signal
J6 input 3	Selected sensor power	GND	Signal

Each terminal pole has two solder legs; they are one electrical contact. J2/J3 are connected in parallel for bus continuation, with no onboard CAN ground terminal. Common ground comes through robot power. Use twisted H/L wiring; keep branches short. Strip 8 mm using the conductor range specified by Weidmüller for the exact terminal. Press the white release button for removal and for conductor types requiring it. Provide cable strain relief independently of the terminal.

Both CAN connectors are marked **L(G)** for CAN Low / green and **H(Y)** for CAN High / yellow on the component side, following WPILib's FRC CAN wiring convention. The four front labels share one horizontal baseline. The underside carries only the sugarCANe branding artwork; use the top labels and this guide for connections.

Feed J1 from appropriately fused robot distribution. F1 is secondary board protection. Do not backfeed J4-J6 power pins, parallel the two sensor rails, or apply external power to the programming rail. The board is not galvanically isolated.

Protection covers sensor-supply shorts to **ground**. A direct **12 V-to-5 V cross-connection can damage U4 and 5 V sensors**: U4's OUT pin has a 7 V absolute maximum. Software voltage detection does not guarantee it can act before damage. This follows the TPS2553-1 ratings; reverse-voltage blocking is not a 12 V tolerance rating.

Sensor power and software bias

The factory leaves JP101/JP201/JP301 unpopulated for hand soldering. Fit the specified 2 mm headers, then leave caps off for initial bring-up. With power disconnected, fit ONE cap across 1-2 for 5 V or 2-3 for 12 V. No cap disconnects port power. Connecting all three power pins shorts the rails.

The headers are centered at X=7, 23 and 39 mm, Y=60.4 mm. Pins 1-2-3 run left to right from the component side; pin 1 is square. Above each header, the front labels show 5V on the left and 12V on the right. V1/V2/V3 are the selected sensor-power outputs, GND is the return, and IN1/IN2/IN3 are signal inputs. There are no physical input-bias selectors. Set each channel to NPN, PNP, push-pull, float detection or keeper in software; see input modes.

Use Harwin M22-2010305 headers and Sullins SPN02SVEN-RC closed-top caps (2 mm pitch; not 2.54 mm). Caps are friction-retained, not mechanically latched. After voltage/functional verification, support retention with a small removable bead of electronics-grade neutral-cure silicone between the cap exterior and header body, clear of the contacts. This is a proposed retention method; qualify it for your robot vibration/impact environment and enclosure. Allow full cure before use and inspect seating during service. Do not claim these are vibration-qualified merely because there is no slider. Bridge JP2 only at a CAN bus end that needs a 120 Ω termination. Leave it open at intermediate nodes; normally there are two terminations on the entire bus.

Examples: select `npn` for a sinking output (default active LOW), `pnp` for a sourcing output (active HIGH), or `push-pull` for an actively driven logic signal. NPN pull-up current is at most approximately 3.5 mA. Both NPN and PNP compatibility budgets allow 100 µA OFF leakage. No onboard 3.3 V sensor-power output is provided; the signal pull-up is not a sensor supply.

Design targets: 0-30 V positive input, LOW ≤1.2 V / HIGH ≥2.4 V, common-ground offset budget ±0.1 V. Do not use as an analog input, quadrature encoder, IO-Link port or continuous negative-voltage input. Sampling is 1 ms with 1-100 ms debounce; short pulses can be missed. Sensor power budgets are shared, 300 mA per

rail. U4 independently limits 5 V; U9 limits the protected 12 V output at about 357 mA nominal (roughly 314-401 mA tolerance screen). Firmware latches a failed rail off and does not retry automatically. Remove the fault, then issue addressed CAN `power-reset` for the affected rail, or cycle board power. ADC dividers measure both rails; CAN service reports millivolts and enable/good/fault states. These are supply measurements, not sensor analog inputs or current measurements. A fault on either rail makes the health-checked driver input unavailable. See protocol.

Representative industrial sensors

Regulated 12 V is a suitable nominal supply for the following families, subject to exact model and cable checks. Connect the selected output only; insulate unused complementary output wires separately. Brown/blue/black is common supply/ground/output coloring, but follow the specific datasheet rather than assuming colors.

Family	Published limits	Rev H setup / limitation
Omron E3Z three-wire NPN/PNP	12-24 V ±10%; typically 30 mA maximum, through-beam pair 35 mA; residual ≤1 V below 10 mA	Power 2-3 for 12 V. NPN or PNP software mode. Residual voltage fits the LOW target at this board's small load
Banner QS18 standard opposed/diffuse/retroreflective variants	10-30 V; <25 mA excluding output load; standard outputs <1 V at 10 mA	12 V supply is in range. Select NPN or PNP bias to match model; use one output. Account for its 100 ms power-up delay
Banner QS18 DVS/DL/FF variants	NPN residual specified up to 1.5 V at 10 mA	This exceeds the board's design LOW target. Prefer a compatible PNP variant or qualify actual NPN residual at the ~3.5 mA pullup load; do not assume all QS18 NPN models work

Sources: Omron E3Z specifications, Banner QS18 manual. PNP OFF leakage must also meet the input pulldown budget. Observe manufacturer cable/fusing requirements. Two-wire current-loop, AC, analog, IO-Link and safety/OSSD sensors are not supported as interchangeable three-wire inputs.

12 V is specified at the board, not guaranteed at the remote cable end under every transient. The analytical DC divider/reference budget is approximately 11.64-12.38 V before dynamic/distribution allowances; check ≥10.8 V at an E3Z during startup and worst battery/load conditions. Three 30 mA sensors are comfortably within the 300 mA shared design target, but include emitters, indicator loads, cable losses and inrush. The board's 100 ms rail qualification is not proof that every sensor has completed its own startup.

Programming header

J8 is a recessed female **2×5, 2.54 mm through-hole socket**, centered at (24, 6.30) mm. Its electrical mapping is custom, not an ARM/STDC ribbon pinout. Use male-ended mapped leads or a correctly wired 2×5 male plug. In the component-side view, the upper row is **2, 4, 6, 8, 10** left to right at Y=5.03 mm; the lower row is **1, 3, 5, 7, 9** at Y=7.57 mm. Pin 1 is square at (18.92, 7.57) mm. Hole diameter is 1.10 mm. The ten electrical pin assignments and hole positions are unchanged from the previous male header.

J8 pin	Signal	Connection
1	NRST	Probe reset; connect for factory programming
2	MCU UART RX	3.3 V UART adapter TX
3	MCU UART TX	UART adapter RX
4	SWCLK	SWD clock
5	SWDIO	SWD data
6	+3V3	Probe VTref sense **only**, board output
7	GND	Probe ground
8	GND	UART ground
9	BOOT0	Normally leave open; onboard pulldown

J8 pin	Signal	Connection
10	GND	Ground

Power the board through J1 during programming. Leave adapter/probe power disconnected; pin 6 is a voltage-reference input to the probe, not a power input to SC12. Leave BOOT0 open. One reusable ST-Link and five mapped wires program all ten boards. See status and low-cost programming for the exact probe connections and LED legend. Remove programming wires for robot operation.

Disabling a rail opens its supply switch; it does not actively discharge the output. Stored charge decays through connected loads and the voltage divider. `power-reset` clears the protection latch and re-enables the rail; it does not guarantee a particular power-off voltage or reset interval for every sensor.

Rev H design envelope: at most 100 µF total external capacitance on the shared 12 V rail, and 22 µF on the shared 5 V rail, at 300 mA per rail. The startup calculation includes +20% capacitance and the current-limit minimum; the 470 µF simulation is a typical diagnostic beyond that envelope. Qualify nonlinear sensor inrush separately.

5 Software input modes

Source: docs/INPUT_MODES.md

Each input has its own electronically controlled pull-up and pull-down; there are no input-bias jumpers or solder selectors. Select the mode in the desktop setup tool, CAN service CLI, or Java/C++ runtime API. This is an independent design with CANdi-like input choices; CTRE Phoenix and GrappleHook are not compatible protocols.

Mode in setup tool	Electrical behavior	Default active condition	Typical use
`npn`	Pull toward 3.3 V through a diode and 1 kΩ	LOW	NPN / open collector sinking output
`pnp`	Pull toward ground through 10 kΩ	HIGH	PNP sourcing output
`push-pull`	Both bias switches off	HIGH	Actively driven 3.3 V, 5 V or industrial logic
`float`	Alternate pull-up/down to classify the source	DRIVEN	Slow contact / high-impedance diagnostics
`keeper`	Bias toward the last sampled level; starts LOW	HIGH	Hold a released line near its last logic level

Active condition is independently selectable as `high`, `low`, `driven`, `floating`, `not-high`, or `not-low`. Inversion is applied afterward, then debounce. Normally leave the inversion mask at zero when using these active conditions. Firmware defaults to push-pull/HIGH on all channels; it does not guess a sensor type.

Wiring and electrical limits

Power, ground and signal remain the three sensor-terminal contacts. Choose 5 V or 12 V sensor power with the separate removable power cap, with board power disconnected. These power caps and CAN termination are not input-bias settings. Keep a common signal ground.

The pull-up uses the internal **3.3 V rail regardless of sensor supply**. Its series diode blocks most current from a 5-30 V signal toward the MCU rail. The voltage-rated pull-down MOSFET switches a 10 kΩ resistor. The LM339LV comparator, protected divider and hysteresis still receive the field signal; no field-voltage connection goes directly to an STM32 GPIO. This remains digital sensing, not analog acquisition.

Targets: positive 0-30 V input, LOW ≤1.2 V, HIGH ≥2.4 V, ±0.1 V ground-offset allowance. The compatibility budget allows 100 µA OFF leakage for NPN or PNP sensors. NPN LOW residual must remain ≤1.2 V at up to approximately 3.5 mA sink current. Verify the exact sensor datasheet. A nominal 5 V push-pull signal works with bias off. A 3.3 V open-drain output may use NPN mode if it tolerates the board's pull-up while the sensor is unpowered; otherwise use an external pull-up to its own supply with onboard bias off.

Both switches default off through external gate resistors during reset and in the bootloader. Firmware applies saved modes during startup and suppresses measurements until input/rail settling completes. Removing a power cap does not disconnect SIGNAL; an enabled pull-up can inject current into an unpowered sensor. Set push-pull/bias-off before servicing an externally connected unpowered device. Reverse diode leakage is finite: the 10 kΩ 3.3 V rail bleed budgets about 0.61 V from the three bias diodes alone using the BAT46W 60°C specification. Charged sensor-rail capacitors can temporarily add current through the ADC dividers. This is not isolation or a whole-board unpowered-voltage guarantee.

Floating detection and counts

Float mode waits at least 2 ms after each bias change. Two matching full pull-up/down pairs are required, so initial classification needs approximately 8 ms, plus configured debounce for subsequent logical changes. Input-level telemetry reports HIGH, LOW, FLOATING or UNKNOWN (unsettled). Drivers mark logical inputs unavailable while any channel is unsettled or either rail is unhealthy.

Probe transitions never enter the logical edge counter. Counters track debounced inactive-to-active transitions. They pause during rail faults and maintenance; startup/reconfiguration re-baselines without inventing an edge. Use float mode for slow signals, and qualify the real cable capacitance, noise and sensor

leakage. Fast changes during a probe pair can be misclassified; fixed NPN/PNP/push-pull modes are preferable for ordinary sensing. This is not an encoder interface.

High impedance can mean a disconnected cable, an open contact, or the normal inactive state of an NPN/PNP output. Float detection cannot distinguish these causes. Keeper mode biases a released wire; it does not retain a value across reset or report a disconnection.

Save an example configuration

From `current/`, after discovery and physical identification:

```
python host/can_service.py --bridge localhost:5810 configure \
  --uid UID --device 17 --invert 0 --debounce 2 \
  --modes npn,pnp,push-pull
```

Replace `UID` with the discovered 24-digit hexadecimal serial. The tool stages all three settings, commits with the ID/settings record, resets, and verifies persistent readback. If `--modes` is omitted, existing saved input settings are preserved. Optional `--active low,high,high` overrides the defaults. The desktop window offers the same choices independently for each channel.

Java `configureInputsRuntime(InputMode[], ActiveWhen[])` and C++ `ConfigureInputsRuntime(...)` apply volatile settings. Read back `inputState()` / `ReadInputState()` to verify the configuration word and classification; runtime CAN commands have no acknowledgment and do not write flash. Use CAN service to persist. See wire format, service and drivers.

Engineering checks and simulations do not qualify electrical behavior. Bench-test all modes across supply, temperature, cable length, hot-plug, sensor startup and robot vibration before deployment. See validation.

6 Status and programming

Source: docs/STATUS_AND_PROGRAMMING.md

Rev H has one RGB LED, D10, in the open area on the right, above the right-hand mounting hole. A silkscreen circle and RGB label identify it. It runs from the internal 5 V supply, so it can indicate a sensor-power fault while the protected outputs are off. Sensor-voltage selection remains on the removable 2 mm jumpers.

Reading the indicator

Pattern	Meaning
Green, one short pulse per second	Configured, rails qualified, telemetry enabled, and a CAN transmission acknowledged within 500 ms
Amber, one short pulse per second	Waiting: unconfigured, startup, maintenance, or telemetry disabled; also the initial CAN connection grace period
Red, two short pulses per second	Latched sensor-power fault or clock-start failure; inspect CAN diagnostics and rail measurements
Red, equal on/off every 250 ms	CAN initialization/error-passive/bus-off fault, or no successful transmission after the initial one-second grace period
Blue, one 200 ms pulse per second	Bootloader, including CAN update/recovery
Magenta, rapid 50 ms on/off	Three-second identify request; application power/CAN faults take priority
Dark or frozen color	Check board power, reset and firmware; this alone does not prove the outputs are off

A CAN acknowledgment proves that another node accepted the frame at the bus level; it does not prove the robot application consumed it. Green does not certify sensor type, jumper voltage, robot enable state or healthy external wiring. There are no per-input LEDs. This is SC12's own status scheme and CAN protocol.

Enclosure location

Coordinates use the component-side view, top-left PCB corner as (0,0). D10 is centered at **X=48 mm, Y=39.5 mm**: 4 mm from the right edge and 7 mm above mounting hole H2. Its nominal body is 1.5 × 1.5 × 0.65 mm. A **6 mm diameter area** centered on D10 is reserved against other component courtyards for a future circular light duct. C75 sits just above this area and remains connected by a short local supply trace.

For a first printed enclosure, use a duct with **6 mm maximum outside diameter and 4 mm inside diameter**. These are nominal starting dimensions, not a tolerance-qualified enclosure. The duct must hang from the case, without loading the PCB or LED. Start its lower face **0.2-0.5 mm above the LED**, or nominally **0.85-1.15 mm above the top PCB surface**. Maintain the separate mounting-hardware clearance at H2; do not combine the duct with a screw boss.

The LED should be visible from above, but brightness and viewing angle have not been measured on hardware. Keep a hollow duct short and its opening unobstructed; a translucent window or light-pipe insert can make the case indicator easier to see from an angle. Check all RGB patterns through the actual print under workshop and competition lighting before finalizing the case. The CAD render shows the package without illumination.

J8 is centered at (24,6.30) mm, with a 13.10 × 5.00 mm insulator and approximately 8.5 mm total height above the PCB. Its tails extend about 1.6 mm beyond a 1.6 mm PCB. Allow access through a removable lid and space for plugged leads during servicing. The completed STEP model and mechanical drawing include both parts. All assembly remains on top.

One programmer for the ten-board batch

Hand solder the **XFCN PM254V-12-10P-H85 recessed female 2×5, 2.54 mm J8 socket** included in the hand-solder BOM. Its contacts sit inside the insulating body instead of projecting above it. The estimated undiscounted price is $0.1158 each at ten, approximately $0.044 more per board than the former male header. No special pogo cable is required. Keep one short five-wire harness and reuse it across all boards. Disconnect it for robot use.

Use an existing ST-Link with external-target SWD, reset and voltage sensing, or a Nucleo's detachable/external ST-Link according to that specific board's manual. If buying a programmer, the genuine STLINK-V3MINIE was listed at about **$26 on 2026-09-09**, before shipping/tax and USB cable. That is a one-time bench tool, about $2.60 per board spread over ten, plus a small wire/adapter allowance.

The V3MINIE's CN2 edge pads are **2 mm**, and its supplied ribbon is **1.27 mm STDC14**: neither directly plugs into this 2.54 mm socket. Make one mapped adapter, or solder strain-relieved flywires to CN2 following the probe manual. Verify every connection unpowered. Use ordinary male 2.54 mm leads at the SC12 socket end. A single 2×5 male plug with only the required wires populated is easier to orient consistently; match the pin-one triangle and the custom mapping.

Signal	SC12 J8	STLINK-V3MINIE CN2 edge pad	Alternative CN4 STDC14 pin
NRST	1	9	12
SWCLK	4	4	6
SWDIO	5	3	4
3.3 V target reference, sense only	6	10	3
GND	7	6	5

Use only one probe connector path. CN2 uses pads on both faces; identify positions from ST UM2910, connector tables and board views. J8's full UART/BOOT0 mapping and odd/even orientation are in WIRING.md. J8 is a custom signal assignment, not an ARM ribbon-compatible socket.

Power SC12 through J1, with sensor jumper caps removed for first bring-up. Connect J8.6 only to target-voltage sensing; never feed 3.3 V or 5 V from the programmer into it. Leave J8.9 BOOT0 open. With short leads, start SWD at 1 MHz or lower:

```
openocd -f interface/stlink.cfg -f target/stm32f0x.cfg \
  -c 'adapter speed 1000' -f tools/program_factory.tcl
```

Run from `current/`. The script programs the matching combined factory image, verifies it, and sets/verifies bootloader write protection. Power cycle and follow the fixture checks. The first installation needs SWD; later application updates, CAN ID and sensor modes use CAN service. Bootloader or option-byte recovery still requires SWD. USB-C is needed on the external programmer, not on each SC12 board.

Circuit and verification basis

PB3 drives an SN74AHCT1G125 buffer, which converts 3.3 V data to the LED's 5 V logic domain. A 10 kΩ pulldown holds its input low on reset; a 330 Ω series resistor and local 100 nF capacitors support the short board-level link. OPSCO SK6805-EC15 uses GRB data at 800 kbit/s. The buffer is necessary because this LED's minimum HIGH level is 0.78 of its supply; directly driving it with 3.3 V would lack guaranteed margin.

The timer/DMA implementation preserves foreground input sampling and CAN service. Compiled firmware tests exercise the emitted data, timing registers, transfer errors, CAN acknowledgment tracking and application status states. They do not measure real LED waveforms, optical brightness, or reset behavior on silicon. Bench-check colors, timing, CAN unplug/recovery, rail faults and boot-to-application transitions before robot use. The LED may hold its last color until the next valid word if the MCU resets or stops.

Sources: OPSCO SK6805-EC15 A/1 datasheet, TI SN74AHCT1G125, STM32F042 datasheet, ST RM0091.

7 Assembly

Source: docs/ASSEMBLY.md

Factory: fit only the 129 SMT references in Factory_BOM.csv using CPL_JLCPCB.csv. User: buy the MCU and ten through-hole parts and three closed-top caps.

Use the exact BOM, assembly drawing, interactive BOM and placement files. Reference labels are in these drawings; dense placement does not leave room to print every reference legibly on the actual top silkscreen.

1. Order the 52 × 74 mm, four-layer, 1.6 mm board with 1 oz finished copper target. Use the exported Gerbers and separate plated/nonplated drill files. Have the fabricator review its actual stackup, annular rings, mask registration and minimum features. Use Standard PCBA with the new high-reflow C1 and its specified dwell limits; see factory instructions. ENIG is retained for the fine-pitch SMT pads; hand solder the service header after reflow.

2. **Factory:** check bare-board continuity and orientation. Apply paste only to the factory-fitted SMT pads. U5 must be empty and unpasted; use the factory-specific paste layer from the order ZIP. A 0.10 mm stencil is a starting point; the assembler must review release and the VSSOP pad apertures. Through-hole header pads, the termination solder bridge and the test point deliberately have no paste.

3. **Factory:** place 0805/1206/1210 passives, diodes and ICs. Verify C1 polarity, TVS/diode cathodes, pin 1 on each IC and the exact latch-off TPS2553DBVR-1 suffix. U8 TPS61085DGKR uses the supplied 0.65 mm-pitch VSSOP footprint. Confirm D4 cathode orientation and U4 TPS2553DBVR-1 and U9 TPS259620DDAR, including the exposed ground pad and six adjacent thermal vias. D10 must be OPSCO SK6805-EC15 with the supplied pin mapping; U10 is SN74AHCT1G125DBVR. Do not substitute a different addressable LED footprint. There are no components on the bottom.

4. **User, after delivery:** inspect joints and measure for supply shorts. Install U5 first, before the tall connectors. Buy exact STM32F042K6T6 / LQFP32 from DigiKey. Match its pin-one mark to the MCU orientation map, align all 32 legs, tack opposite corners, recheck alignment, then flux/solder the remaining leads with an iron. Wick bridges, clean and inspect all leads under magnification. There is no central exposed pad. A full-board stencil cannot sit over the already-fitted SMT parts; whole-board hot-plate reflow is not required. Hand solder the six horizontal Weidmüller terminals after SMT reflow as the user assembly step. Fit J8, the recessed female 2×5 2.54 mm service socket, with pin 1 at the square pad. Then fit JP101/JP201/JP301: 2 mm headers on the top side, straight and fully seated, with pin 1 at the square pad. Use a removable cap temporarily as an alignment aid with the board unpowered; remove it afterward. Fill both plated holes per terminal pole. Support the board without loading fresh SMT joints. The terminal pins extend below the board.

5. Leave all selectors open for initial bring-up. Supply J1 from a current-limited bench supply, starting at 12 V and a modest current limit. Confirm input polarity, protected input, approximately 5.07 V internal, 3.3 V logic and approximately 12 V at the raw BOOST_12V reservoir. With a blank MCU, protected sensor rails should remain disabled; do not expect normal LED or CAN operation until programming. Sensor headers remain unpowered while power selectors are open.

6. Program the combined factory image through the custom J8 mapping, set/verify boot protection, power cycle and configure a unique CAN ID using CAN service. Follow the fixture procedure. Save each input mode in software and verify readback. Then power off, choose each port's supply and termination settings, inspect one-cap-per-header placement, and check that 5 V and 12 V are not shorted.

7. Perform the qualification checks before connecting a robot harness. Begin with a known load and known logic source, then the actual sensor/cable.

Use M3 hardware at (3.5,4) and (48.5,46.5) mm. Holes are 3.2 mm NPTH. The copper keepout has at least 3.30 mm radius; use hardware whose contact area stays within a 6.4 mm diameter, with insulating washers/spacers as appropriate. Account for component bodies, connector overhang and protruding pins. Do not clamp the PCB between conductive surfaces. The mechanical drawing gives nominal geometry, not a fitted enclosure.

U5 is the only SMT part assigned to the user. The other ICs, VSSOP, RGB LED and passives remain factory assembled. Inspect the user-soldered 32 MCU leads and 49 through-hole joints before power-up. See jumper retention and vibration qualification in WIRING.md.

8 Firmware

Source: firmware/README.md

Bare-metal STM32F042K6T6 firmware for three digital inputs. Build allocation is recorded by the current linker map/build report; stack depth has not been measured. The final two 1 KiB flash pages hold redundant 32-byte versioned configuration with CRC and commit-last writes.

Rev H uses **hardware identity 8** and **protocol 4**. It has a separate image tag and configuration magic so D2 images/settings are rejected. Use the matching drivers and software input modes. The separate power caps select sensor voltage.

CAN service is implemented, physically unqualified: full-UID discovery, identify LED, persistent ID/inversion/debounce and verified uploads. See CAN setup/update, robot bridge and qualification plan.

From `current/`:

```
source tools/env.sh
make -C firmware
bash scripts/test_firmware.sh
```

Outputs: `build/factory.hex` for initial SWD installation; `build/application.sc12` for CAN updates; separate bootloader/application ELF, BIN and MAP files. `sensor_can.hex`/ELF are application-only at 0x08002000. Flash: 8 KiB bootloader, 21 KiB app, 1 KiB manifest, 2 KiB configuration. Both linkers reserve the first 256 SRAM bytes and a 1 KiB minimum stack. The MCU controls six input-bias gates and two sensor-rail enables, and reads divided 5 V/12 V supplies through ADC channels 8/9. Per-input LEDs are not fitted. The 8 MHz external crystal and PLL generate 48 MHz; bxCAN uses 16 time quanta with an 87.5% sample point at 1 Mbit/s. Auto bus-off recovery is enabled; automatic transmit retry is disabled.

Power through J1 from a bench supply. Use the custom 2.54 mm header wiring: VTref sense→6, SWDIO→5, SWCLK→4, ground→7, reset→1. Leave probe power disconnected.

```
openocd -f interface/stlink.cfg -f target/stm32f0x.cfg \
  -c 'adapter speed 1000' \
  -f tools/program_factory.tcl
```

This command has not been exercised on physical Rev H hardware. Verify MCU identity, supply and flash readback. Normal application programming should preserve the two configuration pages; full-chip erase removes them. Version-1/2 configuration is rejected.

UART: adapter TX→J8.2, RX→J8.3, ground→J8.8; 3.3 V, 115200 baud, 8N1; adapter power disconnected.

```
python host/sensor_can.py info --port /dev/ttyUSB0
python host/sensor_can.py configure --port /dev/ttyUSB0 \
  --device 17 --invert 1 --debounce 2
```

Device IDs 0-63 must be unique in this Team Use address space. Invert is a 3-bit mask; debounce is 1-100 ms. UART `SET id invert debounce_ms` and UID-targeted CAN service save persist and reset. Legacy CAN runtime configuration remains volatile. Unconfigured boards accept discovery/service but suppress normal telemetry. Power-cycle after initial SWD programming and verify boot protection before CAN updates. **Both sensor rails default off during reset/programming and remain off in the bootloader.** The application enables 5 V after 10 ms and 12 V after 60 ms; hardware current limiting remains independent of software. On Rev H, leave the caps off JP101/JP201/JP301 to disconnect port power during initial bring-up; configure them only with input power disconnected. Input bias is electronic; CAN termination remains a solder setting. See assembly.

The independent watchdog is serviced only after the full 1 ms foreground cycle. A >20 ms foreground delay, clock failure, hard fault or PVD low-voltage detection puts CAN into standby; the watchdog provides reset recovery. The RGB LED indicates ready (green pulse), waiting/configuration (amber pulse), power/clock fault (red double pulse), CAN failure (red blink), bootloader (blue pulse), or identify (magenta blink). Ready requires a recent successful CAN transmission. See the complete legend and timing. PA3/PA4 read active-low 12 V/5 V fault outputs with internal pullups. PB0/PB1 measure divided 5 V/12 V with the calibrated internal reference. PA15 enables 5 V; PB7 enables the Q3/Q2 12 V stage. PB5/PB6/PB4 drive input pull-downs 1/2/3. A qualified fault disables the rail until explicit CAN reset or a board restart. ADC failure disables both rails. Startup allows 200 ms, then an undervoltage persisting 20 ms latches off; overvoltage or asserted switch fault trips immediately. Healthy rails need 100 ms continuous qualification before counts resume. See `src/power.c` for exact thresholds and unit tests.

Measured health bands are 11.1-12.9 V and 4.55-5.50 V; low-voltage latch thresholds are 10.8 V and 4.4 V. These bands include ADC/reference/divider error and are separate from the calibrated-meter acceptance limits. Overvoltage trips remain 13.0 V and 5.6 V. The electrical report records the measurement-error budget.

Broadcast Disable ID 0 stops telemetry and aborts pending transmissions. Local sampling and application-managed sensor power continue. An explicit RESUME UART command or CAN resume command re-enables telemetry; reads never do this automatically. See protocol v4. Host tests cover firmware logic and mocked peripheral boundaries, not actual oscillator startup, electrical timing or STM32 peripheral behavior.

The RGB serializer uses PB3 AF2/TIM2_CH2 and DMA1 channel 2 (TIM2_UP), at 800 kbit/s. It emits GRB, most-significant bit first, with a low reset interval exceeding 200 µs. Timer preloading and two trailing zero slots finish the final bit before DMA stops. The foreground never busy-waits or masks interrupts for LED transmission. A DMA error or transfer lasting beyond 2 ms is stopped and retried later; boot-to-app handoff stops DMA before SRAM is reused. R31 holds the buffer input low during reset. An addressable LED can retain its last color through an MCU reset or crash; interpret repeating patterns, not a frozen color, as firmware activity.

9 CAN protocol

Source: docs/CAN_PROTOCOL.md

1 Mbit/s classic CAN, extended 29-bit identifiers, 8-byte telemetry. This is an original Team Use protocol, not CTRE Phoenix. Device type=10 (Miscellaneous), manufacturer=8 (Team Use), unique device ID=0-63. Address = `(10 << 24) | (8 << 16) | (API << 6) | device`.

The APIs differ from A/B/B-2; Rev C shares these API numbers but uses protocol v2, which Rev H drivers reject. The version and capability checks prevent drivers from interpreting the wrong revision's payload. Provision each board by full-UID CAN service or 3.3 V UART. An unconfigured board suppresses telemetry but answers service requests; its sensor power hardware is still enabled.

All multibyte fields are little endian. Masks use bits 0-2 for inputs 1-3. Unsupported versions, reserved bits, wrong lengths or stale frames must be reported as unavailable.

API	Period	Eight-byte payload
0x120 status	10 ms	version=4, sequence, debounced input mask, raw mask, channels=3, capabilities=0x7F, flags, reset flags
0x121 counts 1/2	100 ms	two uint32 rising-edge totals
0x122 count 3	100 ms	uint32 total, uint32 reserved=0
0x130 diagnostics	100 ms	uint32 transmit drops, TEC byte, REC byte, LEC byte, CAN flags byte
0x132 input state	Same as status	version=4, valid mask, floating mask, classified HIGH mask, uint32 input configuration
0x131 identity	1000 ms	4, firmware major=0, minor=9, patch=0, hardware=8, device ID, debounce ms, inversion mask

Status flags: configured=1, clock healthy=2, CAN initialized=4, bus-off=8, qualified 12 V power-good=16, qualified 5 V no-fault=32, all input classifications settled=64. Reset flags: power-on=1, pin reset=2, watchdog=4, software=8. Capabilities bits 0-4 indicate edge counts, runtime configuration, UART persistence, standalone hardware power and broadcast disable. Bit 5 indicates rail-health reporting capability; bit 6 indicates software bias/classification. Capabilities are static; flags bits 4/5 carry live qualified indications. The raw mask is the classified HIGH mask, excluding float probe transitions; check the valid mask in 0x132 before interpreting it. The debounced mask applies the active-state condition, inversion and debounce. Sequence is an unsigned byte and wraps. Counters wrap at 2^32 and clear on reset; runtime configuration preserves totals.

Diagnostic LEC is the STM32 three-bit last-error code (0-7); CAN flags are bus-off=1 and error-passive=2. CAN service INFO 10 reports measured 12 V/5 V in millivolts; INFO 11 reports enabled/good/latched/pending rail masks. ADC calibration and physical accuracy need bench verification. Normal status reports qualified rail health; it is not current telemetry. Raw masks remain diagnostic data during faults.

A bad rail indication clears its flag at the next 1 ms sample; restoration needs 100 consecutive good samples. During faults and recovery, firmware re-baselines input debounce and freezes edge totals, preventing supply collapse/recovery from adding counts. All ports require both rail indications because sensor-power cap positions are not readable. Java/C++ input helpers also require fresh telemetry, configuration, clock health and CAN health and all classifications settled. They cannot detect every failed/disconnected sensor or open cable.

Command API	Payload	Effect
0x220 runtime input config	`53 43 04 invert debounce_lo debounce_hi 00 00`	invert 0-7; debounce 1-100 ms; volatile, does not change ID
0x222 runtime bias	`53 43 04 00` + uint32 configuration	Volatile, re-baselines input classification; read back 0x132
0x221 resume telemetry	`53 43 04 00 00 00 00 00`	Explicitly resume after broadcast disable

Extended arbitration ID **0** is the FRC broadcast disable. Firmware stops telemetry and aborts pending transmit mailboxes immediately on processing it. Sampling and standalone sensor power continue; there are no actuator outputs. Resume requires the explicit command above, UART `RESUME`, or reset. Drivers never resume from a read loop. Integrate explicit resume with a reviewed robot lifecycle policy; this implementation still requires robot-bus testing and does not establish competition approval. See WPILib CAN addressing.

Read timeouts: status 100 ms, counters/diagnostics 350 ms, identity 1500 ms. Counters 1/2 and 3 are separate snapshots. Sampling is 1 kHz; debounce requires a continuously stable logical state for the selected interval. This is not a high-speed pulse counter or encoder.

UART: 115200 baud, 8N1, 3.3 V logic. Commands `INFO`, `HELP`, `SET id invert debounce_ms`, `RESUME`. `SET` saves a version-4 redundant CRC-protected record and resets. The old four-argument command and earlier flash records are rejected; provision Rev H explicitly. No power-control command exists.

Service extension in firmware 0.9.0

Rev H retains protocol 4 with hardware identity 8; use only the matching drivers. APIs 0x300/0x301 at address 0 are reserved for CAN service. Opening maintenance pauses telemetry/counting on participating SC12 nodes; treat samples as unavailable immediately in robot code. Status period can be 10/20/50 ms until reset.

Per-input configuration word

Bits 0-2, 3-5 and 6-8 select channels 1-3 mode: 0=OFF, 1=PULL_HIGH, 2=PULL_LOW, 3=FLOAT_DETECT, 4=BUS_KEEPER. Bits 9-11, 12-14 and 15-17 select each active condition: 0=HIGH, 1=LOW, 2=DRIVEN, 3=FLOATING, 4=NOT_HIGH, 5=NOT_LOW. Bits 18-31 must be zero; other enum values are invalid. Default word zero means push-pull/HIGH on all channels.

Input state 0x132: a cleared valid bit means UNKNOWN regardless of the retained high/floating bit. For a valid channel, floating takes its own state, high means HIGH, otherwise LOW. High/floating masks never overlap. Validity/classification precedes active-state mapping, inversion and debounce. See mode timing and electrical limits.

At the default 10 ms period, telemetry is 231 frames/s; 20/50 ms give 131/71 frames/s. Status and input state are separate snapshots. Read timeouts are 100 ms for both; 350 ms for counts/diagnostics and 1500 ms for identity. Runtime configuration does not reset counters or persist to flash. Measurements settle after a mode change before further edges are counted.

10 CAN setup and updates

Source: docs/CAN_SERVICE.md

Rev H firmware **0.9.0**, bootloader **1.1.0**, service protocol **1**. Normal sensor messages remain protocol **4**, hardware **8**. Implemented and software-tested; actual STM32/robot-controller, power-cut and vibration qualification is pending. This is our own service protocol, not GrappleHook-compatible firmware.

First installation and recovery

Install factory.hex once through SWD using program_factory.tcl. This combines the protected bootloader, relocated application and image manifest. Rev H requires its own factory image and fresh configuration. D/D2 hardware cannot gain the new bias circuit through firmware. The script preserves the two configuration pages and refuses an unexpected device/flash size or an already protected bootloader. Its hardware execution is a pilot qualification gate.

From `current`, with J8 wired per the custom mapping and a separately powered, current-limited board:

```
openocd -f interface/stlink.cfg -f target/stm32f0x.cfg \
  -f tools/program_factory.tcl
```

Power cycle afterward and verify `boot_protected: true` using the tool. Write protection covers flash pages 0-7 (OpenOCD protection blocks 0-1). Updating requires those blocks locked and application/configuration blocks writable. The script does not change readout protection or mass-erase configuration. Do not program `sensor_can.elf` or `.hex` alone as an initial install: these are relocated application-only files without the boot-valid manifest. Do not upload `factory.hex` over CAN.

Normal CAN updates use application.sc12. If interrupted, retry the complete file: there is no old-image rollback or cross-session resume. Invalid images remain in the bootloader. A normal boot gives a 500 ms discovery opportunity; an explicit boot request stays there, and three successive watchdog resets before two healthy application seconds also request recovery. `run` exits a valid bootloader image. SWD remains the recovery path for bootloader/option-byte damage or a nonfunctional CAN connection.

Laptop and robot use

Use Python 3.10+; the robot tunnel transport needs only the standard library. SocketCAN additionally needs `python-can`; configure the adapter for classical CAN at 1 Mbit/s. Tk is required only for the desktop window (`python3-tk` on many Linux distributions).

The robot bridge runs in a background thread alongside robot code. Copy `ServiceBridge.java` and the adapter for your WPILib year from examples/bridge. It listens only on `127.0.0.1:5810`; access it over an authenticated SSH tunnel. The 2026 adapter requires an explicit maintenance-request flag **and** Driver Station disabled state for every transmission. The 2027 alpha-7 adapter uses `RobotState` and an explicit `CANPort`, compiled separately with JDK 25. Choose a classical-CAN-compatible bus; this MCU cannot transmit CAN FD.

Keep the robot disabled, mechanically secure it and keep outputs inhibited in your robot code throughout maintenance. The bridge rejects or disconnects when permission/disabled state is lost; it cannot prevent someone enabling the robot at the Driver Station. No robot program is stopped or redeployed by this tool. Do not expose this unauthenticated TCP framing directly on a network. The sensor protocol itself has no authentication; CRC detects corruption, not a malicious uploader.

Example using a roboRIO reached over USB; use your controller's actual address/account:

```
ssh -N -L 5810:127.0.0.1:5810 lvuser@172.22.11.2
python host/can_service.py --bridge localhost:5810 discover
python host/service_gui.py
```

Copy the desired 24-hex-digit UID from discovery. Replace `UID` below with it:

```
python host/can_service.py --bridge localhost:5810 blink --uid UID
python host/can_service.py --bridge localhost:5810 configure \
  --uid UID --device 17 --invert 0 --debounce 2 --modes npn,pnp,push-pull
python host/can_service.py --bridge localhost:5810 update \
  --uid UID --file firmware/build/application.sc12
```

Direct SocketCAN uses `--channel can0`. Other commands: `info`, `run`, `rate --milliseconds 10|20|50`. Settings persist after save/reset; status period resets to 10 ms. Existing driver timeout is 100 ms. Traffic is 231/131/71 frames/s at 10/20/50 ms; verify bus jitter physically.

Discovery resolves the complete STM32 96-bit UID independently of assigned CAN IDs, including blank and duplicate-ID devices. Blink confirms physical selection. Before a change, the tool reads back all three UID words. ID conflict checks cover **responding SC12 devices only**; a powered-off board or another Team Use product can still have a conflicting address. Use one maintenance client per CAN network. The service APIs must be reserved against other Team Use devices.

Opening a session suppresses normal telemetry on all participating SC12 boards, and freezes/rebaselines input counts. Application rail supervision continues during maintenance; the bootloader keeps both rails disabled. Drivers may retain their last sample until its normal freshness timeout; robot code must treat maintenance as unavailable immediately. Close/lease expiry restores normal sampling/telemetry unless Broadcast Disable previously latched it off. Broadcast Disable cancels maintenance and pending transmissions; explicit resume remains necessary for that latch. Never interpret a missing sample as a false input.

Wire protocol

Classical CAN, extended identifiers, eight data bytes, no RTR. Request ID `0x0A08C000`, response ID `0x0A08C040` (Team Use 8, miscellaneous type 10, APIs 0x300/0x301, address 0). Payload: opcode byte, sequence byte, nonzero little-endian session token (16 bits), little-endian value (32 bits). Response opcode OR 0x80 means success; OR 0xC0 means error. Echo sequence/token and validate the expected value. Errors 1-5 mean argument, state/offset, flash, image, and missing boot protection. No reply means an unmatched session/selection or transport failure.

Operation	Value / result
OPEN 0x01	Magic 0x314D4353; broadcast ACKs are identical; five-second lease
PREFIX0/1/2 0x02-04	Three raw UID words; no ACK; each change clears selection
PROBE 0x05	Prefix length 0-96, MSB first within each UID byte; matching nodes reply 1; only 96 selects
CLOSE 0x06	End session, no ACK
INFO 0x10	Index 0 flags/versions, 1 settings, 2 mode version, 3 upload offset, 4 capacity, 5 image validity, 6-8 UID words, 9 persistent input configuration, 10 rail millivolts, 11 rail states
BLINK 0x11	Zero; three-second identify LED
SAVE 0x12	ID/invert/debounce/reserved bytes; reserved must be zero; explicit save then reset
BOOT / RUN 0x13/14	Zero; enter bootloader / run verified application
RATE 0x15	10, 20 or 50 ms; volatile status period
INPUTS 0x16	Stage validated per-input configuration; commit with SAVE; cleared on new selection/session
POWER 0x17	Application only, selected UID: mask 1=12 V, 2=5 V, 3=both; disable then restart requested rails
SIZE / CRC 0x20/21	Padded application length and CRC32; bootloader only
BEGIN 0x22	Product/layout tag 0x53433138; invalidate/read back manifest before application writes
OFFSET / DATA 0x23/24	Byte offset then four image bytes; ACK next offset after flash readback
FINISH 0x25	Zero; validate complete flash CRC/vectors, commit manifest last, ACK image CRC

INFO index 0 packs service version in bits 0-7, bootloader mode in bit 8, configured in bit 9, boot protection in bit 10, application minor 9 in bits 16-23, and hardware 8 in bits 24-31. Index 1 packs ID/invert/debounce/status-period as bytes; unconfigured ID is 255. Index 2 is major/minor/patch in bits 16/8/0. Index 5 returns 0 or 1.

INFO index 10 packs measured 12 V millivolts in bits 0-15 and 5 V millivolts in bits 16-31. Index 11 packs enabled/good/latched/pending masks in bytes 0/1/2/3; mask bit 0 is 12 V and bit 1 is 5 V. Bootloader replies do not imply powered rails. Millivolt data is ADC-based and needs physical calibration verification.

After removing a sensor-power fault, restart only the required rail:

```
python host/can_service.py --bridge localhost:5810 power-reset --uid UID --rail 12v
```

`--rail 5v` and `--rail both` are also supported. The desktop tool exposes **Reset power fault** and displays both voltages. Reset is explicit; the application does not repeatedly retry a short. Firmware uses 10/60 ms off-to-enable delays and requalifies the rail before reporting usable inputs. A persistent fault latches off again. The outputs have no active discharge, so this clears protection without guaranteeing a complete power cycle of every sensor.

Host discovery walks binary UID prefixes; matching responders send identical presence frames. Prefix frames are paced; UID readback guards selection before writes. Queries retry four times, probes twice; default pacing is at least 6 ms between requests and a 300 ms ACK timeout. Binary discovery can take seconds per device; intermittent connectivity can omit a node, so repeat discovery before commissioning. Uploads send OFFSET and DATA for each four-byte word, bounding throughput and working around flash stalls. Only the immediately previous exact request is cached; offset/data logic also accepts an identical last word without rewriting it. Gaps, changed duplicate words and stale sessions are rejected.

Image and memory layout

Region	Address / size
Write-protected bootloader	0x08000000 / 8 KiB
Application	0x08002000 / 21 KiB
Image manifest	0x08007400 / 1 KiB
Configuration A/B	0x08007800 / 0x08007C00, 1 KiB each
Reserved SRAM	First 256 bytes: 192-byte vector table, validated recovery cookie

Both linkers reserve at least 1 KiB stack, excluding static data; runtime stack high-water measurement remains pending. The Cortex-M0 application copies vectors to SRAM and remaps them; bootloader handoff clears pending interrupts, stops SysTick/CAN and returns clocks to HSI. Compiled-register tests check vector copying, GPIO/CAN and configuration writes against peripheral models; they do not emulate analog effects or real interrupt timing.

The update file uses `<8s6I`: `SC12UPD1`, format 1, hardware 8, application address, padded length, CRC32, packed version, then image bytes. Length is 192-21504 bytes and a multiple of four. Initial SP must be aligned and within application SRAM, and Thumb reset PC inside application executable bounds. Manifest words are magic 0x49324353, size, CRC, product tag, CRC of the first four words, and final commit 0xC035A19E. Boot verifies manifest, vectors and full application CRC. No update command can write bootloader, configuration or option-byte regions.

Evidence and remaining gates

Tests execute the actual portable C engine with simulated flash, including 2,148 halfword erase/program power-cut positions on a two-page image, retries, corrupt files, wrong sizes, gaps, stale sessions, duplicate IDs and running-version verification. Compiled ARM tests exercise register intent and 34 configuration halfword interruption cases. Real TCP tests cover fragmentation and permission revocation with mock CAN. The Python uploader/GUI, bridge and firmware need end-to-end testing on real hardware before robot use. See qualification and unit programming/test fixture.

The `--modes` option and GUI persist independent input modes and active conditions using INFO 9 / INPUTS 0x16. Omitted modes preserve the existing persistent word; a runtime 0x222 change is not implicitly saved. SAVE commits ID, inversion, debounce and staged input settings together. See input modes.

11 Robot integration

Source: examples/README.md

Java 17 and C++ examples target WPILib 2026.2.1. Rev H uses protocol **4**, hardware **8** and firmware **0.9.0**. Use the matching `SensorCan12` and `SensorCan12Codec` files; older revisions are rejected. Provision a unique device ID through CAN service or UART.

Choose each sensor's 5 V / 12 V supply using the physical jumper before operation; the drivers do not select port voltage. See the Rev H setup guide and wiring.

CAN discovery, persistent settings and firmware updates are provided by the new service tools. The robot bridge has separately compiled 2026 and 2027 alpha-7 adapters. Sensor driver classes below remain pinned to 2026.2.1; the 2027 bridge does not port the normal drivers. Physical end-to-end CAN testing is pending.

```
private final SensorCan12 sensorBoard = new SensorCan12(17);
// In periodic(): absent means unavailable or a reported fault.
var detected = sensorBoard.input(1);
if (detected.isPresent()) {
  boolean sensorActive = detected.get();
  // Use a reviewed subsystem policy for this observation.
}
```

C++: `sensorcan12::SensorCan12 board{17};` then `board.ReadInput(1)` returns `std::optional<bool>`. Construct the driver once; an empty result must not silently become a valid “no object” observation. The helper requires fresh status, configured identity state, healthy clock/CAN and both qualified power indications. Raw `status()` / `ReadStatus()` remains available to diagnose faults. Counts are frozen during power faults and wrap at 2^32; compare/reset policy in the robot application.

Reads do not transmit or resume telemetry. `configureRuntime` / `ConfigureRuntime` sends volatile input inversion/debounce; it does not set persistent ID. `resumeTelemetry` / `ResumeTelemetry` is an explicit action after investigating broadcast disable, not a periodic automatic retry. Status timeout is 100 ms; count/diagnostic timeout 350 ms; identity timeout 1500 ms.

The Python host tool supports UART `info`, `configure`, `resume` and passive CAN `monitor`. UART uses 3.3 V levels at 115200 8N1; disconnect adapter power. CAN monitor requires a suitable external CAN interface; a UART adapter is not a CAN interface. Run the following from `current/`:

```
source tools/env.sh
python host/sensor_can.py configure --port /dev/ttyUSB0 --device 17 --invert 1 --debounce 2
python host/sensor_can.py monitor --interface socketcan --channel can0 --device 17
bash scripts/test_drivers.sh
```

Tests cover codec boundaries, all reserved status bits, stale/malformed responses, fault rejection, unsigned counts and explicit-only writes. The C++ example compiles against the actual pinned WPILib headers; Java compiles against pinned WPILib jars. Hardware bus behavior and competition acceptance remain open qualification steps. See protocol, wiring, and validation.

Independent input modes

Use the desktop service tool to persist per-input settings. Runtime Java example (volatile; verify `inputState()` readback):

```
sensor.configureInputsRuntime(
    new SensorCan12Codec.InputMode[] {
        SensorCan12Codec.InputMode.PULL_HIGH,
        SensorCan12Codec.InputMode.PULL_LOW,
        SensorCan12Codec.InputMode.OFF},
    new SensorCan12Codec.ActiveWhen[] {
        SensorCan12Codec.ActiveWhen.LOW,
        SensorCan12Codec.ActiveWhen.HIGH,
        SensorCan12Codec.ActiveWhen.HIGH});
```

C++ exposes `ConfigureInputsRuntime` with `std::array<InputMode,3>` and `std::array<ActiveWhen,3>`, and `ReadInputState()` for classification/configuration. `input()` / `ReadInput()` require fresh, healthy status and all inputs settled. Level helpers return UNKNOWN when the channel valid bit is clear. See mode behavior and electrical limits.

12 Purchasing and factory quote

Source: docs/FACTORY_ORDER.md

Selected build: JLCPCB assembles 129 top-side SMT parts. You buy the original STM32F042K6T6 from DigiKey and solder U5, then fit the ten through-hole parts and three jumper caps. The MCU footprint, four-layer layout and firmware stay the same. The final component review changed C25/C26 to **20 pF / C105173** for crystal startup margin, and U1/L1 to **LMR51430XDDCR / C5185863** plus **SRP7028A-8R2M / C2041815** for buck current headroom. All four use the existing footprints.

Engineering review passed: the final power, input, firmware and four-layer CAD screens are complete. The ZIPs remain REVIEW_ONLY until JLC confirms the specified stack, solder process and inspection, and you approve the quote. Use this package to request that confirmation. No physical prototype has been tested. Results and remaining gates.

1. Buy the parts you will fit

Order the MCU, terminals, voltage headers and caps from **DigiKey**. Order the recessed programming socket from **LCSC**. Ship both orders to yourself in the USA. Do not send the MCU to JLC or start a JLC Global Sourcing order.

Part / supplier link	Where it goes	For 5 boards	For 10 boards
STM32F042K6T6, DigiKey 497-14647-ND	U5 MCU	**7 recommended**	**12 recommended**
Weidmüller 1824420000, 281-2020-ND	J1-J3, two-pole terminals	15	30
Weidmüller 1824430000, 281-2021-ND	J4-J6, three-pole terminals	15	30
Harwin M22-2010305, 952-2274-ND	Three 2 mm voltage headers	15	30
Sullins SPN02SVEN-RC, S9340-ND	Three 2 mm jumper caps	15	30
XFCN PM254V-12-10P-H85, LCSC C492399	J8 recessed 2×5 / 2.54 mm socket	5	10

MCU quantities include **two spare chips**; each board needs one. Other quantities are exact requirements before spares. Use the five-board DigiKey import or ten-board import; both include two spare MCUs. J8 is a separate LCSC purchase. Check each cart's quantity, manufacturer and delivery date.

Buy **K6T6 / LQFP32**, 7 × 7 mm body with visible 0.8 mm-pitch legs. **K6U6TR / QFN does not fit this PCB.** DigiKey displayed 3,071 K6T6 in stock and $3.277 each at 10+ on September 10, 2026 UTC; twelve cost about $39.32 before shipping, tax and cart charges. Its 40-week manufacturer replenishment field is separate from stocked inventory. Stock is not reserved.

Reuse one compatible ST-Link and a mapped five-wire harness. If you need a probe, see the STLINK-V3MINIE and adapter instructions. J8 has a custom pinout; an ordinary ARM ribbon cannot plug in directly. Have a temperature-controlled iron, fine solder, flux, tweezers, wick, magnification, multimeter and current-limited bench supply.

2. Download the matching JLC files

Extract sugarCANe_RevH_JLC_REVIEW_ONLY.zip. Do not upload the whole documentation packet as the Gerber file. Use these three files from that same packet:

JLC upload field	File
PCB Gerber upload	`order/sugarCANe_RevH_Gerbers_REVIEW_ONLY.zip`
BOM	`bom/BOM_JLCPCB.csv`
CPL / pick-and-place	`output/assembly/CPL_JLCPCB.csv`

The BOM contains **54 types / 129 placements**; the CPL contains **129 rows**. **U5 is absent from both.** The Gerber ZIP has a factory-specific top paste layer with all **32 U5 apertures removed**. The other 325 apertures, including U9's exposed-pad windows, are preserved. Copper, solder mask and drills remain the checked native design. Ask CAM to use the supplied stencil geometry without recreating U5 openings.

The complete BOM, assembly drawing and 3D renders show the **finished board, including your MCU and connectors**. They are reference material, not instructions to add factory placements. Do not upload `BOM_grouped.csv`, `positions_all_top.csv` or `positions_smd_top.csv` as the JLC assembly selection.

3. Configure JLCPCB

Start at JLCPCB PCB quotation, upload the Gerber ZIP, and choose PCB assembly. Interface labels can change; use these requirements:

Setting	Select / require
Finished quantity	Your chosen **5 or 10 assembled boards**; match PCB and assembly quantities
PCB	**52 × 74 mm**, **four layers**, FR-4, **1.6 mm**
Copper	**1 oz / 35 µm nominal on all four layers**, including both inner layers
Finish	**ENIG**
Mask / silk	Green mask, white silkscreen on both sides; retain bottom artwork
Assembly	**Standard PCBA, top side only**, 129 placements
Panel / rails	Factory adds rails/fiducials and returns separated 52 × 74 mm boards
Electrical test	Confirm bare-PCB open/short electrical test
Stencil	Top only, nominal **0.10 mm**, supplied apertures; **no paste at U5**
Programming / functional test	You perform these after installing U5

D10's exact RGB LED requires Standard PCBA. Do not select Economic. The board is narrower than the published Standard minimum; have JLC confirm its panel/rail arrangement. Default rails alone must not be assumed to satisfy it. Request a symmetric four-layer stack with outer-to-inner dielectric ≤0.25 mm and 35 µm copper on every layer; do not accept 0.5 oz inner copper without another electrical review.

Upload and process BOM/CPL. **All 54 listed MPNs must be selected**, with no unexpected shortfall or unmatched parts. U5 and the ten through-hole parts are intentionally omitted; every other SMT part must be present, including C1, D10 and U9. Advancing past an unmatched part can omit it from assembly. Official matching guidance.

Inspect the placement preview against top_assembly.pdf, particularly IC pin 1 and diode/capacitor polarity. CPL coordinates are millimetres from PCB lower-left, X right and Y up, with native KiCad counterclockwise rotation. The factory must reconcile its part-library orientations; do not guess a blanket 90° correction.

4. Copy these notes into the PCBA order

sugarCANe SC12 Rev H. TOP assembly only: 129 placements / 54 exact MPN types from the supplied BOM/CPL. Leave U5 (STM32F042K6T6) unpopulated and unpasted; customer hand solders it. Use the supplied F_Paste Gerber with U5 apertures removed. Leave J1-J6, J8 and JP101/JP201/JP301 unpopulated; leave JP2 termination bridge open. No part substitutions or added placements. Fit C1, D10 and every other BOM/CPL reference. U1 is LMR51430XDDCR; L1 is SRP7028A-8R2M (8.2 uH). X-ray U9 TPS259620DDAR's exposed ground/thermal pad on every board and provide the report; confirm D10 underside-joint inspection coverage. Confirm 35 µm copper on ALL FOUR layers, symmetric 1.6 mm stack, outer-to-inner dielectric ≤0.25 mm, Standard-PCBA panel/rails and depanelization to 52 × 74 mm. Confirm component-compatible reflow/moisture handling, including D10 MSL 4 and C1 limits below. No firmware programming or powered functional test requested; U5 is absent. Confirm production and tracked express delivery timing to the USA before payment.

C1 reflow limits: Panasonic EEEFT1H470AP, profile (5): case-top peak ≤260°C; at/above 250°C ≤5 s, 230°C ≤30 s, 217°C ≤40 s and 200°C ≤70 s; at most two cycles. D10 is exact OPSCO SK6805-EC15; U8 is TPS61085DGKR; U4 is **TPS2553DBVR-1**; U9 is TPS259620DDAR. The `-1` latch-off suffix is required. Confirm the factory process and any X-ray charges in the quotation. Process evidence · JLC targeted X-ray requests.

5. Review the quote and delivery before paying

1. Review the passing engineering results, obtain the factory confirmations below and issue the approved manufacturing package before paying; the current package remains REVIEW_ONLY.

2. Verify quantities, all 54 selected types, U5/stencil omission, stack and component orientations. Files do not reserve inventory.

3. Obtain confirmation of reflow handling, panelization and U9/D10 inspection. Review any CAM changes before acceptance.

4. Check the **complete** price: PCB, parts, assembly/setup/stencil, panel/rails, X-ray, packing, express freight and tax/duty. Ten-board parts are about **$28.06 per board** with conservative JLC attrition and your MCU, before two optional spare MCUs and those additional charges. This is not a final assembled-board price.

5. Your last arrival date is **October 10, 2026, USA**. Use October 9 as the practical delivery target. Request a dispatch estimate and destination-specific tracked express quote; allow time for soldering/testing. Chinese holiday closures and customs can affect timing. Displayed production duration is not the arrival date. Order the separate DigiKey/LCSC parts in time too.

JLC public stock covers all **54 factory types for ten boards**, including displayed allowances, in the September 10, 2026, 07:22 UTC checks. U9 has 25 available and L2 has 39, so recheck at checkout. Allocation and delivery are unconfirmed. Full audit.

6. Finish the delivered boards

Start with one board. Inspect it unpowered, then solder **U5 before the tall connectors** so you can access its legs. Use the MCU orientation map and assembly procedure.

Use **flux and a soldering iron**: align pin 1, tack opposite corner leads, recheck alignment, solder the remaining leads, and wick any bridges. This LQFP has no hidden central joint. Avoid prolonged repeated heating; clean and inspect all 32 leads under magnification. A full-board stencil cannot sit over fitted components. Your hot plate is optional rework equipment; this plan does not require reheating the factory-fitted LED and capacitor.

Fit terminals, service socket and voltage headers. Keep all caps removed and JP2 open. Check for bridges and supply shorts, then follow the current-limited fixture procedure. A blank MCU will not provide normal status or sensor-power operation yet.

Program the matching combined `firmware/build/factory.hex` over SWD; it includes bootloader and application. From `current/`, with OpenOCD installed and the verified J8 wiring:

```
openocd -f interface/stlink.cfg -f target/stm32f0x.cfg \
  -c 'adapter speed 1000' -f tools/program_factory.tcl
```

Power through J1. J8.6 is **target-voltage sense only**; never feed power from the probe into it. Verify programming/protection, power-cycle and set a unique CAN ID. With power disconnected, select each port's voltage: **pins 1-2 = 5 V; pins 2-3 = 12 V; no cap = disconnected sensor supply**. The middle pin is sensor-power output, not ground. Bridge JP2 only at a CAN bus end to connect the existing 120 Ω resistor; do not add a second resistor there.

Complete first-board acceptance with a known load before connecting sensors or a robot harness, then repeat recorded checks for the batch. Initial programming needs SWD; later application updates and settings can use CAN.

Printable guide · Factory package · Hand-solder BOM

13 Qualification

Source: docs/VALIDATION.md

No Rev H prototype has been assembled, powered or tested. The automated evidence checks the design, software and manufacturing package. Physical acceptance remains a separate step.

Automated evidence

Native KiCad ERC/DRC, explicit model/schematic/PCB parity, assembly constraints and independent reconstruction of all four Gerber copper layers check the delivered geometry. Firmware tests exercise the application state machine, CAN servicing, ADC scaling and register-level GPIO behavior; host and Java/C++ tests check protocol handling. Mocked hardware does not prove physical STM32 or CAN behavior.

Electrical calculations cover input thresholds, sensor leakage/residual-voltage margins, divider stress, capacitor derating, converter current budgets and output current-limit tolerances. The rail-reading budget reserves ±2.5% gain and ±90 mV offset at the sensor rail; firmware health bands include this uncertainty. The averaged boost loop screen covers 768 CCM corners. See the release report for the exact retained results and hashes.

Manufacturer-model transient screen

Current run status: PASS transient screening. All nine power cases and both complete 40 ms startup traces pass on the final BOM. Startup settles at 11.975-11.980 V, with 12.054 V peak protected output and 1.544 A peak inductor current. The two startup runs differ by at most 4.15 mV on internal 5 V, 14.23 mV on raw boost, 10.37 mV on protected 12 V and 0.524 mA in peak inductor current. The hot-short half-timestep repeat differs by at most 0.77 mV and 0.39 mA. Full results · Current waveforms. Factory confirmation and physical qualification remain separate gates.

The harness maps the fitted TPS61085 and TPS259620 pins/components into original TI models, with the Diodes B240A rectifier model. Results retain every scenario, criterion and generated netlist; current Rev H waveforms show the responses. A failed criterion returns nonzero and prevents the release validation command from passing.

Scenario	Engineering question
Low-input combined startup, 66 µF raw bank, 100 µF external and 300 mA	Does controlled load connection avoid the former input-limiter startup failure?
30→310→410→0 mA boost load	Does the converter regulate at normal load and the output limiter's upper current bound?
Running short and explicit reset	Does output protection limit current and recover when commanded?
Startup into a short and explicit reset	Does the circuit remain bounded without requiring a successful first startup?
470 µF external capacitance with 300 mA	Does a larger capacitive load start in the standalone eFuse model?
5 V startup, short and enable reset with concurrent 1.3 A load on internal 5 V	Does the unchanged TPS2553 latch-off branch work with its new GPIO enable?
Half-timestep repeat of the running 12 V short	Are the fault/reset results numerically stable away from imposed switching edges?
Sustained short, removal, then enable cycle	Does the thermal-latch model remain off until deliberate recovery?
Reduced current-limit surrogate, 120 µF external and 300 mA	Does the supported capacitance plus 20% start with minimum charging-current headroom?

Capacitance stress cases use constant nameplate values; ceramic DC-bias reduction makes actual capacitance voltage-dependent. This is a conservative capacitive-load screen, not a guaranteed sensor-capacitance rating. Actual sensors can have constant-power or nonlinear startup loads.

The short tests connect a supply output to ground. They do not establish survival when the 12 V and 5 V outputs are joined: that can exceed U4's 7 V OUT absolute maximum and damage 5 V devices. Cross-rail wiring and externally powering output pins remain unsupported; do not interpret current limiting or software rail monitoring as protection against these faults.

The reduced-limit case uses a 2.92 kΩ test resistor in the typical eFuse model, producing approximately 314.4 mA, below the calculated 314.7 mA minimum for the fitted 2.61 kΩ resistor. This deliberately harder charging condition reaches 11.4 V in about 93 ms with no model fault. It supplements the analytical startup budget; it does not represent a full silicon or temperature corner model, and does not change the BOM.

The upstream supply is idealized with 120 mΩ series resistance, 25.92 µF internal capacitance and 450 mA concurrent logic/RGB/5 V load. The exact LMR51430 has no compatible public switching model listed on its TI product page. Its control loop/current limit, PCB/cable inductance, robot brownouts, temperature and production corners are not simulated. The default-off transistor stage is represented by an idealized enable switch; its physical startup behavior needs the separate bench check below. The eFuse thermal timing comes from its behavioral model, not a board thermal solution.

The normal boost case uses fixed derated C60/reservoir values. Silicon models are typical 27°C; inductor DCR is 32 mΩ and inductance is linear below the specified 8 A/20%-drop domain. D5 uses the vendor diode model; D6 uses an approximate static TVS clamp. This does not establish fast fault overshoot or reverse-power survival.

PSpice adaptations are syntax-only: namespacing, duplicate PARAMS removal and Boolean XOR translation. The combined startup uses KLU/Gear with `xtrtol=2`, a 50 ns maximum step and a 1 fF numerical node shunt. A complete repeat uses `xtrtol=7` and a 25 ns maximum step. Both retain relative tolerance 0.003 and the same circuit/models, and must pass all electrical criteria. Over the full 40 ms, the comparison requires voltage differences below 50 mV and peak-inductor-current difference below 20 mA. These are solver controls, not component changes.

The original `xtrtol=1` combined run stopped numerically at 28.971 ms during the transition into regulation. It and other incomplete solver diagnostics are excluded from passing evidence; an incomplete trace cannot establish circuit operation or hardware failure. Standalone enable conductance ramps over 1 µs; a discontinuous ideal enable switch also produced a numerical timestep failure. The finite transition keeps the same 2 Ω final shunt. Zero-supply operating-point warnings are accepted only after successful transient-OP fallback and full completion. No semiconductor parameter is tuned to obtain a pass. Results have not been correlated with PSpice or a physical board. Rev F's superseded failure reports remain in its immutable archive.

Firmware monitor waveform replay

The replay screen feeds four completed SPICE traces into the actual `power.c`, compiled with ASan/UBSan, at all four combinations of ±2.5% ADC gain and ±90 mV rail offset. It also subtracts the worst current-layout branch drop at 300 mA. All sixteen cases pass: slow capacitive startup does not cause a false trip; running 12 V short causes a software latch after 20 ms; the 5 V hardware-fault signal causes a latch at the 33 ms sample. Neither rail automatically restarts in the replay, and the other rail stays enabled.

The 12 V trace is aligned to the firmware's 60 ms enable; the running-short case holds a pre-short regulated sample until startup grace ends. Samples are 1 ms apart and the last trace sample is held after its end. This is one-way replay, not coupled hardware/firmware simulation: shutdown commands do not alter the recorded waveform. The 470 µF case remains an extra diagnostic beyond the supported 100 µF external capacitance.

Sensor-input transient screen

Ten input-model runs, including two half-timestep repeats, exercise the three native divider/filter/feedback circuits together. The waveforms cover 3.3/5/24 V push-pull, 1.2/2.4 V logic limits and 30 V input, NPN/PNP with 100 µA OFF leakage, floating and driven probes, hysteresis, and comparator supply cycling. Cable capacitances are 0.1, 1 and 10 nF; probe checks sample before the firmware's 2 ms settling interval expires.

TI links its LM393LV channel model from the LM339LV product page. Three instances represent the used channels; native resistor/capacitor values and shared reference are retained. The model uses typical comparator behavior. The bias FETs, diode and sensors use explicit resistance, voltage-drop and leakage abstractions, not transistor-level models. The independent 4,096 static threshold corners remain necessary; these transients do not establish worst-temperature behavior, arbitrary sensor compatibility, MCU sampling of short pulses, or whole-board backfeed safety. The report records model/source hashes, decks, completed trace lengths and numerical-repeat differences.

Prototype gates before production release

Gate	Measurement / acceptance intent
Assembly	Confirm exact horizontal terminals, pin-to-hole fit, polarity, U8 VSSOP joints, D4 polarity, U9 TPS259620DDAR and exposed-pad soldering, selector isolation and no solder shorts
Staged power	No sensors first; current-limited 12 V source, verify 5 V/3.3 V before full operation, then regulated 12 V and low-load input current
Regulation	Sweep 6-22 V with 0-300 mA on each rail separately and together; verify 12 V remains within the selected sensors' supply limits at their cable ends
Cascaded startup	Scope V5_INTERNAL, BOOST_12V, BOOST_SS, SENSOR_12V and 3V3 during slow/fast input ramps, repeated starts and brownouts. Confirm boost startup/inrush does not falsely latch U9 or collapse U1; review operation below 6 V
Transients	6↔14↔22 V line steps; 0↔300 mA load steps; representative sensor startup/inrush and cable lengths; check ringing/undershoot on both cascaded converters
Stability	Measure loop response or justified equivalent transient qualification on the cascaded 5 V buck / 12 V boost, including CCM and light-load/DCM modes; target >45° phase and >10 dB gain margin where measured
Faults	Short each sensor supply and each port; verify current-limit behavior, hardware current limiting, firmware latching and explicit CAN reset, firmware flags, no spurious counts and safe temperatures. Measure peak current and output-capacitor discharge energy before U9 disconnects; verify cold-start into a short and hot-plug loads
Electrical protection	Reverse input, realistic protected-branch fault energy, fuse clearing/I^2t, inductive transients and TVS stress using suitable test equipment. Not automotive load-dump qualified
Temperature	Both rails fully loaded at 6/12/22 V, intended ambient and later enclosure; measure U8/U1/L1/L2/Q1/fuse/terminal and MCU temperatures, hot-start/restart and surface safety
Inputs	0-30 V PNP/NPN/push-pull thresholds, off-state leakage/residual voltage, ground offset, sensor disconnection, wiring transients and all intended sensor models
CAN/firmware	Real roboRIO bus at 1 Mbit/s with correct end termination; no normal telemetry before provisioning, unique IDs, bus-off, clock faults, watchdog, broadcast disable, explicit resume, stale data and power-fault handling
EMC/mechanical	Motor-noise environment, ESD at terminals, cable retention, vibration, mounting insulation and programming-adapter fit; repeat after enclosure design
Production process	Approved stackup/CAM, stencil/profile, supplier alternates, test fixture, serial/lot records, programming/provisioning and final functional limits

FRC integration

Use a single appropriate protected power-distribution branch and wiring sized for its protection. The 0451 3 A fuse has a published 300 A interrupt rating at 32 V DC; confirm prospective branch fault current and clearing energy for the actual harness. The onboard fuse is secondary protection, not permission to bypass the robot distribution rules. Sensor grounds are common with the board; isolate conductive sensor cases where required by frame-ground rules. Do not use dedicated radio/roboRIO power outputs as a convenient supply.

The custom Team Use CAN address uses manufacturer 8 and miscellaneous device type 10. The implementation processes broadcast disable ID 0; it has no actuator outputs. It is not a CTRE Phoenix device. Robot inspection and the current season's rules still apply. Commercial CAN product acceptance, assigned identifiers and required ecosystem support are separate outstanding steps; Java/C++ examples alone do not establish approval.

Primary references: 2026 FIRST manual, especially custom-circuit voltage, distribution/wire and frame-ground rules; WPILib CAN addressing and acceptance requirements. Recheck applicable rules for the actual event/season.

Rev H-specific prototype checks

Measure the 5 V/12 V ADC readings against a calibrated meter throughout the supply and load sweep. Check 11:1 divider scaling, VDDA compensation, settling and fault thresholds. Confirm both rails stay disabled during SWD programming, bootloader operation and MCU reset. Remove 3.3 V while raw boost remains energized and verify the Q2/Q3 stage holds the 12 V supply switch off. There is no active output discharge: distinguish residual capacitor charge from a conducting switch, using a known test load and probing EN. Check the six U9 ground thermal vias and exposed-pad solder joint.

At minimum input, test fully loaded startup and simultaneous 5 V / 12 V faults; record shared-rail sag, dropout and reset behavior. The new LMR51430/SRP7028A-8R2M pair passes nine current-limit corners, including the charging corner that exceeded the old buck's minimum valley limit. This calculation does not guarantee uninterrupted operation: source impedance, closed-loop behavior, input dropout and thermal limits remain unmeasured. Confirm the output eFuse nominal 357 mA limit, tolerance bounds, load-capacitance startup and thermal latch behavior. Verify a short on either branch does not cause unsafe shared-supply oscillation. The 768-corner CCM screen uses assumed amplifier/modulator bounds and does not qualify silicon behavior.

Verify the exact 2 mm header and cap fit, both 5 V / 12 V positions and unpowered open state on every channel. Verify no connection between the two rails in either position. Check solder fill on all 49 user-fitted through-hole joints. Fit/retention, vibration and impact testing must include the actual caps, chosen removable retention compound, enclosure and robot harness; these headers have no positive latch. Compare the substituted passive values/ratings against the supplied BOM. Factory BOM and CPL must contain exactly 129 SMT references, with U5 and all ten through-hole components omitted. Inspect the additional 32 user-soldered MCU leads; factory paste must omit U5. Firmware identity is hardware 8 / protocol 4; prior-revision firmware is rejected.

CAN servicing qualification

The service engine and host uploader pass simulated flash cuts/retries; ARM register checks and both bridge adapters compile. Before robot use, measure actual boot/vector/interrupt behavior, option-byte protection, stack use, upload time, CAN load, bus-off recovery and duplicate-ID discovery. Interrupt power during invalidation, every erase/program phase and final commit on physical units. Verify persistent configuration, bootloader recovery, controller disconnect/enable rejection and healthy telemetry after exit. Protocol and evidence; per-unit fixture records. No software simulation closes these physical gates.

Electronic bias qualification

Verify Q101/201/301 PMOS and Q102/202/302 NMOS polarity and correct exact MPNs. Scope all six gate signals during reset, SWD programming, bootloader, watchdog recovery and supply ramps: pull-up gates HIGH and pull-down gates LOW mean both biases off. Verify mixed NPN/PNP/push-pull operation and nonvolatile readback independently on all three channels.

Sweep external signal 0-30 V in every mode. Measure inactive sensor leakage margins, active residual voltage and pull-up sink current (~3.5 mA maximum). Test 3.3 V and 5 V push-pull, unpowered sensors and unpowered board backfeed. Qualify the BAT46W leakage/rail-bleed budget over the actual operating temperature; current calculation uses its 60°C reverse-leakage specification, not a zero-backfeed claim.

Test floating detection with open contacts, NPN and PNP inactive outputs, long cables and added capacitance. Verify UNKNOWN during inconsistent probe pairs, no probe-generated edge counts, and one debounced edge on actual active-state transitions. Check keeper retention while released and LOW initialization after reset. Verify fixed input modes for faster signals; float detection can alias changing sources.

Rev H design envelope: at most 100 µF total external capacitance on the shared 12 V rail, and 22 µF on the shared 5 V rail, at 300 mA per rail. The startup calculation includes +20% capacitance and the current-limit minimum; the 470 µF simulation is a typical diagnostic beyond that envelope. Qualify nonlinear sensor inrush separately.

Rev H RGB and service header

The circuit regression compares H against G and records the status indicator, its buffer/passives, the service-header replacement and the explicitly reviewed JLC substitutions. C25/C26 use 20 pF to improve calculated crystal startup margin. Power component values and sensor inputs are unchanged. Compiled ARM tests check TIM2 update DMA mapping, both 8/48 MHz pulse settings, GRB encoding, reset gaps, bounded transfer-error recovery and CAN acknowledgment freshness. The application harness checks status during normal, unplugged-CAN, initialization, clock and power-fault cases. These tests emulate CPU/register boundaries, not the actual timer/DMA silicon or emitted optical waveform.

Follow the fixture checks and status/programming guide on the first assembled unit. Verify LED package orientation and 5 V buffer margin, reset/brownout transitions, no sampling interruption, the five-wire programming harness, and the enclosure optical opening. The nominal STEP model is not physical-fit evidence.

14 Unit fixture

Source: docs/FIXTURE.md

Build a reusable fixture around the documented 2.54 mm J8 header and five mapped wires, ST-Link, a separately current-limited J1 supply, accessible rail test contacts, CAN adapter and three independently driven input fixtures. This document and scripts are ready for the pilot; no physical fixture or unit test has been performed.

Leave voltage caps absent for initial programming. Keep the SWD probe's voltage pin as target reference only. Program/verify the combined image and set boot write protection with program_factory.tcl, then power cycle and verify protection over CAN. Save the programmer log, factory image SHA-256, component lot and full MCU UID. The setup checks STM32F04x device ID 0x445, 32 KiB flash and WRP pages 0-7 (OpenOCD protection blocks 0-1, not 0-7), consistent with the OpenOCD STM32 driver. Qualify the exact probe/OpenOCD version on the first unit.

Configure a unique ID through the service tool. Record internal 5 V, logic 3.3 V, protected sensor 5 V and regulated 12 V with a calibrated meter; acceptance windows are 4.8-5.3, 3.15-3.45, 4.65-5.45 and 11.4-12.6 V. These generous unit-screening windows do not replace rail regulation/load/temperature qualification. Power off before fitting selector caps. Exercise each input low and high using the approved bias and compatible test signal; observe fresh protocol-v4 status and count transitions. Verify telemetry after closing maintenance. Power cycle again and verify saved ID/inversion/debounce and all three input modes/active conditions.

Copy fixture_measurements.template.json, enter measured voltages and set each check true only after actually observing it. Record live identity/protection plus those results:

```
python host/fixture_record.py --channel can0 --uid UID \
  --measurements measured-unit.json --operator NAME --lot LOT \
  --output unit-UID.json
```

Use `--bridge localhost:5810` for a robot tunnel. An incomplete/failed record exits nonzero. Existing output files are never overwritten. The JSON stores operator/lot/time/UID/settings and a hash of the supplied measurements; keep it with the programming log. It does not independently measure voltages or validate operator assertions.

Design qualification still needs 6/12/22 V input tests, both rails at load, shorts, temperatures, representative sensors/cables, vibration, actual flash power cuts, bus-off/controller disconnect and boot/interrupt timing. See VALIDATION.md. Complete the pilot before calling units production-ready or quoting a qualified manufacturing yield.

Exercise mixed software modes and verify floating detection without false counts using INPUT_MODES.md. Unsettled inputs must not appear as valid robot measurements.

Retained power qualification requires recorded checks for the U9 exposed-pad joint, rails off in bootloader/reset, ADC readback against a calibrated meter, and each rail shorted then recovered through addressed CAN reset. During first-article qualification also remove MCU power while raw boost is present; verify Q2/Q3 holds the 12 V output off. Use controlled test equipment and load fixtures for these checks; do not count an unperformed test as passed.

Rev H also requires checking all RGB colors against the legend, CAN unplug/recovery, boot/update/identify and both rail-fault indications. Measure DIN low/high levels, 800 kbit/s bit timing and reset-low interval >200 µs, including startup on 8 MHz fallback and boot-to-application handoff. Verify input sampling and CAN service continue during LED updates. Confirm header orientation and fit, remove the programming leads, and inspect LED visibility through the proposed case/light pipe. Record these observations with the unit log; no such physical test has yet been performed.

15 Connectors

Source: docs/CONNECTORS.md

References	Exact MPN	Series description	DigiKey
J1-J3	**1824420000**	LSF-SMT 3.50/02/90 3.5SN BK TU	281-2020-ND
J4-J6	**1824430000**	LSF-SMT 3.50/03/90 3.5SN BK TU	281-2021-ND

Manufacturer pages: 2-pole, 3-pole. `/90` gives horizontal wire entry parallel to the PCB. `/135` is the angled version and is not used. Phoenix PTSM is not fitted.

These are through-hole/THR parts despite “SMT” in the family name. Pitch is 3.5 mm, with two solder legs per pole and 7 mm row spacing. PCB drills are 1.1 mm. Housing widths are 7.7 mm / 11.2 mm. Manufacturer CAD is 14.3 mm deep; the footprint conservatively reserves 14.75 mm. Height is 8.5 mm above the mounting surface; nominal pin tips reach 3.5 mm below that surface (about 1.9 mm beyond a 1.6 mm PCB).

Current renders use the official STEP solids with a rigid translation (-2.13,-12.625,-3.5) mm into the footprint origin. They are not approximate connector boxes. Display colors are assigned by the project. Original files, transformed files, URLs, hashes, pin/drill checks and nominal bounds are in vendor_connectors.json. Script `import_connector_models.py` checks all six placed pad patterns and the two vendor models. Do not regenerate models merely to change routing: STEP exporters can alter file headers without changing geometry.

Top terminals face outward at the power/CAN edge; three sensor terminals face outward at the opposite edge. The board and nominal terminal bodies occupy a conservative 52 × 82.1 mm envelope; add room for cable bend, strain relief and button access. Renders do not validate conductor retention, tool clearance in an enclosure, molding tolerances or actual solder fit.

Rev H voltage headers

JP101/JP201/JP301 use Harwin M22-2010305: 2 mm pitch, 0.50 mm square pins, 6 × 2 mm body, nominal 5 mm height above the PCB and 3.4 mm tails. The project's 2 mm header footprint uses 0.85 mm drills in 1.35 mm pads, retaining a 0.25 mm nominal copper annulus, and a 0.25 mm nominal body courtyard margin. Pads 1-3 run left-to-right at Y=60.4 mm; pin-1 positions are X=5, 21 and 37 mm.

The Harwin drawing recommends 0.80 ±0.05 mm holes. The retained footprint adds 0.05 mm nominal diameter for hand assembly: with JLCPCB's normal −0.08/+0.13 mm hole tolerance, the minimum becomes 0.77 mm, above Harwin's 0.75 mm minimum. This trades a slightly larger solder gap for insertion clearance; verify finished holes and solder fill on the first article.

The Sullins SPN02SVEN-RC drawing specifies a closed top, 2 mm pitch, 0.50 mm square mating pins and nominal 4 × 2 × 4.92 mm body. Contact height is 2.45 mm from the entry, within the header’s 3.5 mm mating pin length. Nominal mating geometry agrees; physical fit and retention remain untested. Caps are delivered loose and not included in the assembled-board STEP model. Use the finished cap envelope for enclosure access/clearance and allow room to remove them.

The generic local header STEP represents the nominal pin/body geometry, not manufacturer-qualified tolerances. Vendor connector hashes continue to cover the unchanged six terminal models. See wiring for one-cap-per-header configuration and vibration retention qualification.

J8 is a hand-fitted XFCN PM254V-12-10P-H85 recessed female socket, 2×5 at 2.54 mm pitch, with 1.10 mm plated holes and 1.70 mm pads. The matching nominal model and the full custom pin map are in status and programming and wiring.

Edge clearance and visible markings

The bottom resistor R101 now lies horizontally between the first two sensor terminals: its closest pad is 2.20 mm from the board edge, previously 0.575 mm. R103 and R203 moved into the gap beside the third terminal, with 2.05 mm pad-to-edge clearance, previously 0.80 mm. F1, D102, R22 and R109 also moved inward. R104 has 1.98 mm clearance; the two small parts above J8 have 1.70 mm. The compact buck feedback pair R1/R2 retains its original 1.30 mm pad clearance and quiet local routing. These distances are to copper pads, not component bodies.

The board remains 52 × 74 mm, with all components on top. Keep an insulating enclosure around the board and underside solder tails. The recessed debug contacts reduce accidental bridging from above; the socket is unkeyed and is not sealed against metal debris. Remove the programming cable for robot use.

The power headers sit at Y=60.4 mm. Their 5V/12V labels share a baseline above the pins at Y=58.8 mm. The specified Sullins SPN02SVEN-RC cap has a nominal 4 × 2 mm footprint; both fitted positions leave the labels above it exposed. Front legends identify VIN, CAN L(G)/H(Y), termination, DEBUG, each 5V/12V selection, and V1/V2/V3, GND and IN1/IN2/IN3 at the sensor terminals. Both CAN pairs share a horizontal label baseline: G means green, Y means yellow; see wiring. The underside is reserved for the large sugarCANe wordmark and pixel-cane artwork, with no connection or debug text. Use the top labels and wiring guide. Select voltages with power off; fit one cap per header.

The two bare pads beside TERM are JP2, the optional CAN termination solder bridge. Closing it enables the onboard 120 Ω resistor across CAN H/L; leave it open unless this board needs to terminate a bus end. The single round gold pad near VIN is TP1, connected to GND for a meter or test probe. Both features intentionally remain unpopulated.

The cylindrical part near the left edge is C1, a factory-fitted Panasonic EEEFT1H470AP, 47 µF / 50 V. Its smaller can is nominally 6.3 mm diameter and 5.8 mm tall. The center remains (5,21) mm, leaving 1.85 mm nominal can-to-edge clearance, and it remains below the 8.5 mm terminals. The reviewed size-D lands and high-temperature reflow profile replace the previous 8 mm FK capacitor. Protect this exposed area from impacts and cable forces, with enclosure clearance and cable strain relief. Physical vibration/impact qualification remains necessary; this is the standard part, not a vibration-proof variant.

Layout refinement

The outline now has **1.5 mm corner radii**, within the same 52 × 74 mm dimensions. Mounting-hole centers, connector positions, service socket and the LED duct space stay fixed. All six input-bias transistors share the same Y=57.15 mm row; R4 aligns with the neighboring support parts. Native DRC and the independent Gerber check use the rounded contour, including copper-to-edge clearance around the corners.

LaserCAN's board informed the rounded outline and consistent grouping. Its approximately 25 × 25 mm board places components on both sides and uses a different connector and sensor arrangement. SC12 retains top-side assembly, hand-solderable terminals, three external input circuits and two protected sensor supplies. The refinement changes no purchased parts or functions; it does not claim a lower factory quote. No LaserCAN circuit or library files were copied into the design.

16 Cost reduction

Source: docs/COST_REDUCTION.md

The selected build uses **JLCPCB for 129 SMT placements** and **DigiKey for the user-soldered STM32F042K6T6**. The ten-board parts estimate is **$27.18 per board before SMT attrition**, or about **$28.06** with conservative JLC allowances. Two optional spare MCUs add about **$6.55 per batch**. PCB, factory assembly, inspection, shipping and taxes are additional. The 07:22 UTC catalog refresh changed the earlier parts estimate by about $0.10 per board; the buck/inductor correction itself adds $0.62 per board.

Ten-board basis	Parts per board
JLC factory parts before attrition	$15.20
User-fitted MCU, ten through-hole parts and three caps	$11.98
Combined before attrition	**$27.18**
JLC factory parts including conservative allowance	$16.08
Combined including conservative allowance	**$28.06**

The earlier all-factory plan used a provisional $2.4981 MCU price despite zero JLC stock. DigiKey's observed ten-piece price is $3.277. Moving U5 to user assembly raises the parts estimate about $0.78 per board, and removes one factory feeder ($1.53 per order at the posted Standard rate). It avoids the separate JLC MCU procurement step. Actual savings in procurement fees or lead time require a quote.

Boards	Parts purchasing per board before attrition	Lot parts purchasing
1	$34.33	$34.33
5	$30.44	$152.21
10	$27.18	$271.79
25	$25.06	$626.48

These figures exclude optional spares and factory charges, and do not establish supported assembly quantities or available stock for every lot. The JLC check covers ten boards. The five/ten-board DigiKey import files include two extra MCUs, so their cart totals differ from the bare BOM arithmetic. Other parts may have minimum packs; see cost_detail.csv.

Standard PCBA is required by the exact D10 LED. Posted setup, stencil, 54 loading types and two fixtures total **$132.81 per order**. Placement, X-ray, panel/rails, handling, PCB, freight and tax/duty are extra. The final quote may apply other minimums or fixtures. JLC fee basis · partial scenarios.

The board remains 52 × 74 mm, four copper layers, with all parts on top. No protection components or electrical functions were removed by the assembly-scope change. User soldering totals 32 MCU leads and 49 through-hole joints. Reuse one programmer and mapped harness across the batch; programmer details.

JLC factory stock was checked September 10, 2026, 07:22 UTC; all 54 types covered ten boards with the displayed allowances. Separate supplier links for the six hand-purchased types were also checked. Prices and availability are observations, not reserved stock or a complete quotation. Final engineering screens pass; factory stack/process/inspection confirmation and delivery review remain pending. Ordering guide · sourcing audit.

17 Tools

Source: TOOLS.md

Current workflow: final pre-order simulation and component review are authorized. Run the full electrical/firmware/CAD checks below. A review-only package remains available while any simulation or factory qualification is unresolved.

Run the pre-fabrication review after refreshing the native board and electrical evidence: `python scripts/check_order_readiness.py`. Its `--verify-report` mode checks retained hashes and recalculates routed-copper voltage drop. It does not reroute the PCB or perform physical testing.

The saved native KiCad schematic/PCB are authoritative. Circuit data, schematic, BOM and PCB must agree. Historical placement/router experiments have been removed; validation never reroutes the board.

Routing must pass the enforced routing rules. `python scripts/check_routing.py` rejects curves, arbitrary angles, redundant segments, offset parallel overlaps, pad overshoots, end-cap-only joins, tiny free jogs, sharp free bends and unreviewed track widths/transitions. Saved filled-zone evidence distinguishes ground-pour entries from free ends. The width policy records the intentional exceptions; the report inventories all routed nets. `python scripts/test_routing.py` checks the rule fixtures. Manufacturing export runs the gate automatically; native dangling tracks/vias are errors. Independent Gerber geometry checks also reject narrow same-net ground gaps and weak ground connections.

The sugarCANe bottom artwork is an outlined SVG under `hardware/artwork/`. `python scripts/build_brand_artwork.py` rebuilds it from the original letter grids and pixel illustration, without a font file; `kicad-python scripts/apply_bottom_artwork.py` applies it only to B.Silkscreen. `finish_markings.py` preserves the artwork-only underside when regenerating labels. See the artwork source and provenance.

Tested toolchain: KiCad 10.0.6 and its matching pcbnew interpreter, Python 3.12, Arm GNU GCC 13.2.1, GNU make, host GCC/G++, Java 17, Java 25 for the WPILib 2027 adapter, Poppler, InteractiveHtmlBom 2.11.2 and pinned Python packages. Native tool binaries and dependency caches are outside Git. PDF generation uses DejaVu Sans/Mono.

```
# Run from current/, after installing native tools.
uv venv --python python3.12 .venv
uv pip sync --python .venv/bin/python tools-requirements.lock
source tools/env.sh
python tools/fetch_driver_dependencies.py
CANDI_JDK25=/path/to/jdk25 bash scripts/validate_release.sh
```

`tools/env.sh` uses this directory's venv, or the repository venv. It prepends `$CANDI_TOOLS_ROOT/bin` (default `$HOME/.local/share/candi-tools/bin`). Expected wrappers include `kicad-cli`, `kicad-python`, `host-gcc`, `host-g++`, `ibom` and Arm compiler commands. `kicad-python` must match the installed KiCad binary bindings. Java 17 can be selected with `CANDI_JDK`; the 2027 test downloads its pinned dependencies.

The validation script refreshes electrical budgets, loop sensitivity, board constraints, firmware/driver/bridge tests, native ERC/DRC/parity, manufacturing exports and independent Gerber geometry/connectivity checks. It compares against the small frozen Rev G contract in `hardware/design/rev_g_baseline.json`, without depending on an expanded historical checkout.

After validated changes, refresh documents and inspect the results:

```
bash scripts/render_board.sh
python scripts/compare_assembly_cost.py
python scripts/build_documents.py
python scripts/build_handbook.py
python scripts/build_handbook.py --order-guide
python scripts/build_mcu_map.py
python scripts/review_documents.py
# Inspect final renders/PDFs and update reports/visual_review.json with their hashes.
python scripts/write_validation_summary.py
python scripts/build_release.py
python scripts/verify_archive.py
```

`build_documents.py` creates the factory packet and Gerber ZIP after checking current release and power-review hashes. `build_release.py` also requires a completed visual-review record matching the final artifacts, then writes the source/review archive and manifest. `verify_archive.py` extracts it, verifies all hashes and reruns native checks. The manifest excludes itself and its enclosing ZIP from its own hashes. Review PDF page bounds and rendered pages; automated text bounds cannot detect every overlap.

Optional desktop setup preview: `python scripts/check_gui.py` with a working display, Tkinter and Pillow. It uses simulated data and performs no CAN I/O. `scripts/generate_schematic.py` rebuilds native sheets from `hardware/design/circuit.py`; PCB edits remain a separate CAD operation. `import_connector_models.py` is only for restoring/reviewing exact vendor CAD and transforms, followed by fresh STEP/render exports.

Compiler objects, downloaded Java dependencies, test runners, caches and scratch renderings are ignored. The retained firmware third-party subset supports STM32F042 and GCC; see its provenance. All checks are engineering evidence; physical qualification remains required.

JLC review package with unresolved release gates

`bom/jlc_sourcing.json` records the reviewed public catalog observation, manufacturer/MPN/code matches, six earlier passive substitutions, the C1 reflow/footprint review, MCU procurement instructions, price tiers and stock allowances. `check_jlc_sourcing.py` audits that snapshot against the actual BOM/model; it does not query or reserve live inventory. Recheck the linked JLC listings before purchase.

```
python hardware/design/circuit.py
python scripts/build_bom.py
python scripts/estimate_cost.py
python scripts/check_component_margins.py
python scripts/check_jlc_sourcing.py
kicad-python scripts/check_variant.py
kicad-python scripts/check_board_constraints.py
bash scripts/export_manufacturing.sh
kicad-python scripts/export_board_manifest.py
python scripts/build_factory_files.py
python scripts/build_factory_stencil.py
python scripts/verify_manufacturing.py
python scripts/render_gerber_previews.py
python scripts/verify_gerber_geometry.py
python scripts/verify_gerber_connectivity.py
python scripts/write_validation_summary.py
python scripts/build_documents.py --drawings-only
python scripts/build_handbook.py
python scripts/build_handbook.py --order-guide
python scripts/build_mcu_map.py
python scripts/review_documents.py
# Inspect PDFs; update reports/visual_review.json, then bind the final files.
python scripts/write_validation_summary.py
python scripts/build_jlc_candidate.py
```

The current BOM has full passing power/input/firmware evidence. The summary still holds manufacturing release while factory confirmation is pending. For future unfinished runs, `--allow-incomplete` records that state explicitly; `--defer-simulation` is only for a deliberately deferred review. Neither option establishes a pass. The separate review ZIP builder labels its outputs REVIEW_ONLY; verify the user-purchased MCU and factory quote before issuing the order package.

Sensor-input transient simulation

The separate sensor-input screen is fast and runs as part of release validation:

```
python scripts/check_input_spice.py --jobs 4
python scripts/check_input_spice.py --verify-report
```

It uses the original TI comparator model linked for LM339LV, the three native input/reference networks and explicitly simplified bias/sensor circuits. Ten cases include nominal signals, voltage/leakage limits, floating probes, supply cycling and two half-timestep repeats. Model scope and limitations remain part of the result.

Power transient simulation

The final-order BOM passes all nine power cases plus a complete startup numerical repeat. Both coupled traces reach 40 ms and pass the unchanged electrical and comparison limits. Full results · Waveforms. Manufacturing release remains on HOLD for factory process/stack/inspection confirmation, not unfinished simulation.

The optional native simulator is **ngspice 42 with KLU, XSPICE and PSpice compatibility**. Original manufacturer model files and hashes are retained in `sources/spice/`. The harness reads the maintained circuit's pin connections and component values; it generates simulator-specific copies, decks and raw traces under ignored `tmp/spice/`. The full run includes nine cases plus a complete combined-startup solver repeat; the coupled switching models can take several hours.

```
# From current/, after source tools/env.sh; switching-model runs can take several hours.
python scripts/check_boost_spice.py --jobs 2
# Reuse only completed traces whose deck, model and simulator hashes match.
python scripts/check_boost_spice.py --jobs 2 --reuse
# Fast evidence consistency check (preserves and reports failed findings).
python scripts/check_boost_spice.py --verify-report
# Also require every screening criterion to pass; used by validate_release.sh.
python scripts/check_boost_spice.py --verify-report --require-pass
```

The published summary covers the protected boost stage, not the upstream buck or the whole board. A circuit, budget, source-model or harness change invalidates saved evidence. Each case has an eight-hour wall-time limit; a timeout or incomplete trace cannot pass. `--case NAME` produces separate partial reports and cannot replace the full release result. A failed screening criterion returns a nonzero exit status and remains visible in the report. See the simulation scope. Individual-case outputs remain under ignored `tmp/spice/`; the full run is required for the release report.

Firmware waveform replay and partial checkpoints

`python scripts/check_power_replay.py` replays four completed SPICE traces through the actual `power.c`, compiled with ASan/UBSan, at four ADC-error corners. It reads full-suite traces from `tmp/spice/` by default. `--traces PATH` selects another completed run directory, and `--verify-report` checks retained evidence hashes. The fresh standalone run used `tmp/spice/final-crosscheck-fresh/`. This is one-way replay: firmware shutdown commands do not change the recorded circuit.

While a full power run is incomplete, `python scripts/check_order_readiness.py --allow-incomplete` refreshes native pin and copper-drop evidence with a **HOLD** decision. After replay, `python scripts/write_validation_summary.py --allow-incomplete` publishes the completed checks with the same HOLD. The default commands and factory/release scripts still require full passing power evidence.

18 Sources and notices

Source: docs/SOURCES_AND_LICENSES.md

KiCad library/model notices are retained under `licenses/`, together with Arm CMSIS, ST device-header and WPILib notices. Firmware third-party directories and InteractiveHtmlBom output retain their notices. Original project work has no selected blanket license. Manufacturer documents/CAD remain their owners' material; retaining a file is not a redistribution-license grant.

The sugarCANe underside uses an original pixel-cane illustration and original grid-drawn **sugarCANe** lettering. The vector master and lettering source are retained; no external font is used. No Minecraft asset is included. This branding changes the bottom silkscreen only; SC12 Rev H remains the hardware identity.

Subject	Primary reference
Input fuse interruption / derating	Littelfuse 451/453
Buck and capacitors	TI LMR51430
Regulated 12 V boost	TI TPS61085, exact DGK VSSOP package
Shielded 3.3 µH inductor	Bourns SRP5030TA
Input/output/bias capacitor curve	Samsung CL32B106KBJNNW
Banner supply/residual voltage exceptions	QS18 product manual
Sensor 5 V latch-off limiter	TI TPS2553, -1 variant
Sensor 12 V output eFuse	TI TPS2596, TPS259620 latch-off variant
Comparator/leakage caveats	TI LM339LV
CAN interface / bypass	TI TCAN3413
STM32 pinout, supplies, clock	ST STM32F042
Example industrial output residual levels	Omron E3Z specifications
Terminal geometry and exact MPNs	Weidmüller 1824420000, 1824430000
CAN addressing / broadcast disable	WPILib

Samsung CL31A226KAHNNNE typical characteristic data is retained in `sources/datasheets/samsung_CL31A226_characteristics.json`, inherited from the earlier design's manufacturer retrieval. Typical curves do not establish guaranteed capacitance. DigiKey URLs, exact codes and dated observations are in `bom/catalog_china.json`. Vendor STEP URLs, hashes and transforms are in `hardware/library/3d/vendor_connectors.json`.

Equivalent parts need engineering review; distributor substitution suggestions do not establish electrical/mechanical equivalence. Availability and season rules can change after these observations.

Local source evidence includes the TI TPS61085/TPS2553 datasheets and Samsung CL31A226/CL32B106 characteristic JSON. Typical curves do not establish guaranteed minimum capacitance. U8's footprint and nominal STEP body are from the KiCad 10 Package_SO library. Horizontal connector STEP models are official vendor geometry. The new Samsung CL21B105/CL21B475 substitutions retain nominal ratings; the LDO derating allowance is analytical, not a measured characteristic curve.

Rev H sourcing URLs and tiers are in catalog_china.json. Header/cap geometry links the Harwin and Sullins manufacturer drawings; cost assumptions links the assembly fee and lead-time sources. Stock observations are not factory allocations.

Rev H input bias references: AO3401A, BSS138BK, BAT46W. The local SOT-23 footprint/model comes from installed KiCad libraries under their existing library-license policy. CTRE input-mode API informed feature choices; no CTRE firmware or hardware design is included.

Power transient models: TI TPS61085 SLIM133B, TI TPS2553-1 SLVM426A, TI TPS259620 SLVMD54A, and Diodes B240A. Local provenance and hashes distinguish original archive bytes from the diode model's plaintext transcription. Manufacturer notices remain in each file. They are manufacturer material, not covered by a project license. The harness only creates temporary compatibility adaptations: unique helper names, duplicate `PARAMS:` cleanup, and PSpice Boolean XOR conversion. It uses Gear integration after observing an artificial latch reset with trapezoidal integration. Simulator compatibility is described by the ngspice manual.

Input transient model: TI LM393LV SNOM697A, linked by TI for LM339LV. Input-model provenance records the original file hash and its family-level applicability. The original manufacturer notice is retained. This separate harness uses trapezoidal integration and half-timestep comparisons; the power harness keeps its independently checked Gear settings. Bias FETs and sensor outputs are documented circuit abstractions.

Rev H LED/header: OPSCO SK6805-EC15 A/1, TI SN74AHCT1G125, XFCN PM254V-12-10P-H85. Local geometry provenance describes the project-created nominal LED and female-socket envelopes. ST UM2910 defines the programmer mapping. ST RM0091, DMA table 29, assigns TIM2_UP to DMA1 channel 2; STM32F042 AF2 maps PB3 to TIM2_CH2.

C1’s EEEFT1H470AP replacement uses the Panasonic FT catalog and reflow/land guidance. Its local footprint adapts the KiCad 10.0.6 CP_Elec_6.3x5.8 graphics to manufacturer size-D lands (3.2 × 1.6 mm, 1.8 mm gap). The portable nominal STEP model is the installed KiCad 6.3 × 5.8 mm model, under the retained KiCad library notices. Source and board hashes are in the capacitor review in `bom/jlc_sourcing.json`.